# BAERNSTEIN'S QUASI-NORM MONOTONICITY CONJECTURE FOR POLYNOMIALS WITH UNIMODULAR ZEROS

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*Dedicated to the memory of Professor Albert Baernstein II.*

ABSTRACT. Let $m$ denote the normalized Haar measure on the unit circle $\mathbb{T}$. For $0 < r < \infty$, define $\|f\|_r := \left(\int_{\mathbb{T}} |f|^r \, dm\right)^{1/r}$, with $\|f\|_0$ and $\|f\|_\infty$ interpreted as the geometric mean and the supremum norm, respectively. Set $Q_n(z) = 1 + z^n$. We prove that, for every nonzero polynomial $p$ of degree $n$ whose zeros all lie on $\mathbb{T}$,

$$\frac{\|p\|_s}{\|Q_n\|_s} \le \frac{\|p\|_t}{\|Q_n\|_t}, \qquad 0 \le s \le t \le \infty.$$

This settles Baernstein's quasi-norm monotonicity conjecture. As corollaries, we obtain an $L^r$ extension of Visser's coefficient inequality, the sharp O'Hara–Rodriguez inequality and its higher-power analogues, the Erdős–Szekeres product bound $\left\|\prod_{j=1}^{N}(1 - z^{s_j})\right\|_\infty \ge 2\sqrt{N}$ for all positive integers $s_1, \ldots, s_N$ and Agler–McCarthy's entropy conjecture.

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## 1. Introduction

**1.1. Baernstein's conjecture and its consequences.** Let $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ and $\mathbb{T} = \partial\mathbb{D}$, and let $m$ denote normalized Haar measure on $\mathbb{T}$, given by $\mathrm{d}m(\mathrm{e}^{i\theta}) = \mathrm{d}\theta/(2\pi)$ for $0 \le \theta < 2\pi$. For a nonzero polynomial $f$ and $0 < r < \infty$, write

$$\|f\|_r = \left(\int_{\mathbb{T}} |f|^r \,\mathrm{d}m\right)^{1/r},$$

and use the continuous endpoint conventions

$$\|f\|_0 = \exp \int_{\mathbb{T}} \log|f| \,\mathrm{d}m, \qquad \|f\|_\infty = \operatorname*{ess\,sup}_{\mathbb{T}} |f|.$$

Thus $\|\cdot\|_r$ is a norm for $r \ge 1$, a quasi-norm for $0 < r < 1$, and the Mahler measure when $r = 0$.

The principal aim of this paper is to prove the quasi-norm monotonicity conjecture for polynomials with unimodular zeros, posed by A. Baernstein II in 2008; see [AM21, Conjecture 2.2]. We settle it in the following form.

**Theorem 1.1** (Baernstein's conjecture)**.** *Let $n \ge 1$, and let $p$ be a nonzero complex polynomial of degree $n$ all of whose zeros lie on $\mathbb{T}$. For $Q_n(z) = 1 + z^n$ and $0 \le s \le t \le \infty$,*

$$\frac{\|p\|_s}{\|Q_n\|_s} \le \frac{\|p\|_t}{\|Q_n\|_t}. \tag{1.1}$$

*Equivalently, the function*

$$r \longmapsto \log \frac{\|p\|_r}{\|Q_n\|_r}$$

*is nondecreasing on $[0, \infty]$.*

We first point out several notable consequences of Theorem 1.1; their proofs are given in Section 7. Let

$$p(z) = \sum_{k=0}^{n} a_k z^k = a_n \prod_{j=1}^{n} (z - \zeta_j), \qquad |\zeta_j| = 1,$$

be a nonzero polynomial of degree $n$. For $0 < r < \infty$, put

$$C_r := \|1 + z\|_r, \qquad C_0 := 1, \qquad C_\infty := 2.$$

Since the map $z \mapsto z^n$ preserves normalized Haar measure,

$$\|Q_n\|_r = C_r, \qquad 0 \le r \le \infty.$$

- For every $0 \le s \le t \le \infty$, Theorem 1.1 gives the sharp reverse norm inequality

$$\|p\|_t \ge \frac{C_t}{C_s} \|p\|_s.$$

  Equality is attained by scalar multiples of $z^n - \omega$, where $|\omega| = 1$.

- Taking $s = 0$ in Theorem 1.1 yields the sharp $L^r$ extension of Visser's coefficient inequality

$$|a_0| + |a_n| \le \frac{2}{C_r} \|p\|_r, \qquad 0 \le r \le \infty.$$

The constant $2/C_r$ is best possible. When $r = 0$, equality holds for every such polynomial $p$. For every $r > 0$, equality is attained by

$$p(z) = c(z^n - \omega), \qquad c \neq 0, \quad |\omega| = 1.$$

For $r = \infty$, this recovers Visser's classical coefficient inequality [Vis45, Theorem 3].

- Taking $(s, t) = (2, \infty)$ in Theorem 1.1 gives the sharp O'Hara–Rodriguez inequality

$$\|p\|_\infty^2 \ge 2\|p\|_2^2 = 2\sum_{k=0}^{n} |a_k|^2$$

[OR74, Corollary 1]. More generally, the theorem implies sharp higher-power and even-moment inequalities.

- Let $s_1, \ldots, s_N$ be positive integers, and write $P_N(z) = \prod_{j=1}^{N}(1 - z^{s_j}) = \sum_k b_k z^k$. Taking $(s, t) = (2, \infty)$ in Theorem 1.1 yields

$$\|P_N\|_\infty^2 \ge 2\|P_N\|_2^2 = 2\sum_k |b_k|^2.$$

Combining this inequality with Tang's arithmetic estimate $\sum_k |b_k|^2 \ge 2N$ [Tan26, Theorem 3.2], we recover Tang's recent improvement

$$\|P_N\|_\infty \ge 2\sqrt{N}$$

of the classical Erdős–Szekeres lower bound [ES59].

- Combining the Erdős–Lax inequality [Lax44] with Turán's reverse inequality [Tur40, p. 90, Satz I], one obtains

$$\|p'\|_\infty = \frac{n}{2}\|p\|_\infty.$$

This identity was also established directly for self-inversive polynomials by O'Hara and Rodriguez [OR74, Theorem 1]. Together with Theorem 1.1, it yields the sharp mixed derivative estimate

$$\|p'\|_\infty = \frac{n}{2}\|p\|_\infty \ge \frac{n}{C_r}\|p\|_r, \qquad 0 \le r \le \infty.$$

The constant $n/C_r$ is best possible, with equality attained by

$$p(z) = c(z^n - \omega), \qquad c \neq 0, \quad |\omega| = 1.$$

- Theorem 1.1 further implies the sharp weighted power-sum inequality

$$\sum_{k=1}^{\infty} \frac{|S_k|^2}{k^2} \ge \frac{\pi^2}{6}, \qquad S_k := \sum_{j=1}^{n} \zeta_j^k.$$

The constant $\pi^2/6$ is best possible, and equality holds when the zeros $\zeta_1, \ldots, \zeta_n$ form a regular $n$-gon.

- Finally, for $r > 0$ and $f \in \{p, Q_n\}$, define

$$\mathrm{d}\rho_{f,r} := \frac{|f|^r}{\int_{\mathbb{T}} |f|^r \,\mathrm{d}m} \,\mathrm{d}m.$$

Differentiating the monotonicity asserted in Theorem 1.1 with respect to the exponent yields the relative-entropy inequality

$$\mathsf{D}(\rho_{p,r}\|m) \geq \mathsf{D}(\rho_{Q_n,r}\|m), \qquad r>0.$$

At $r=2$, this is equivalent to the homogeneous entropy inequality proposed by Agler and McCarthy [AM21, Conjecture 1.4] and recently proved, together with its sharp equality statement, by Lei and Zhang [LZ26, Theorem 1.3]. This entropy inequality is closely connected with the celebrated Krzyż conjecture [Krz68], which asserts that if

$$F(z)=\sum_{k=0}^{\infty}\widehat{F}(k)z^k$$

is holomorphic and nonvanishing in $\mathbb{D}$ and satisfies $|F(z)|\leq 1$, then

$$|\widehat{F}(n)| \leq \frac{2}{\mathrm{e}}, \qquad n\geq 1.$$

Agler and McCarthy developed a two-step program toward the Krzyż conjecture: the entropy inequality constitutes the first step, while the second requires a full-degree condition for extremal functions. Under this additional condition, the entropy inequality implies the Krzyż conjecture; see [AM21, Theorem 1.8].

1.2. **Relation to previous work.** The study of integral means of analytic functions can be traced back to Hardy [Har15], who investigated the dependence of boundary means on the radius. Baernstein [Bae74] later introduced the star-function and circular symmetrization methods, which provide a powerful framework for sharp comparisons of integral means. Theorem 1.1 is related to the classical theory of multiplier inequalities for polynomials. The Schur–Szegő composition theorem of de Bruijn and Springer [BS47] gives sharp inequalities for suitable polynomial compositions. Arestov [Are82, Are90] extended their results to a broad class of integral functionals, including quasi-norms, and Pritsker [Pri17] formulated related sharp integral-norm inequalities in terms of polynomial multipliers. These results can imply the endpoint case $s=0$ of Theorem 1.1, but they do not compare two arbitrary positive exponents.

Closely related developments concern sharp derivative estimates under restrictions on the zeros. Motivated by a conjecture of Erdős [Erd40], Lax [Lax44] proved that every polynomial $p$ of degree $n$ having no zeros in $\mathbb{D}$ satisfies

$$\left\|p'\right\|_{\infty} \leq \frac{n}{2}\left\|p\right\|_{\infty},$$

and the constant $n/2$ is sharp. Ankeny–Rivlin [AR55], Boas–Rahman [BR62], and Rahman–Schmeisser [RS88, RS02] subsequently developed related pointwise and integral-norm inequalities under various restrictions on the zeros. These results compare a polynomial with its derivative at a fixed exponent, whereas Theorem 1.1 compares the same polynomial at different exponents.

Two neighboring extremal problems further clarify the scope of the present result. First, the Erdős–Turán theorem [ET50] controls the angular discrepancy of the zeros in terms of a quantitative size parameter of the polynomial. By contrast, the endpoint case of Theorem 1.1 yields only the distribution-free estimate

$$\frac{\|p\|_{\infty}}{|a_n|} \geq 2.$$

Thus any refinement incorporating a stability term that measures the discrepancy of the zero-counting measure from normalized Haar measure would require genuinely new information beyond a further specialization of the exponents in Theorem 1.1. Second, Erdős, Herzog, and Piranian [EHP58, p. 142, Problem 12] asked whether, among monic polynomials of fixed degree, the lemniscate $\{z:|p(z)|=1\}$ has maximal length for $p(z)=z^n-1$. Tao recently proved this conjecture for all sufficiently large degrees [Tao25, Theorem 1.1(iv)].If this lemniscate contains no critical point of $p$, its inverse branches give

$$\operatorname{length}\{z : |p(z)| = 1\} = \int_0^{2\pi} \sum_{p(z)=\mathrm{e}^{it}} \frac{\mathrm{d}t}{|p'(z)|}.$$

This functional depends both on derivatives and on a moving planar level set, whereas Theorem 1.1 controls integral means of $|p|$ on the fixed circle $\mathbb{T}$.

A more recent connection arises from the homogeneous entropy conjecture of Agler and McCarthy [AM21], which was introduced in their study of the Krzyż conjecture. They [AM21, Section 10] observed that Baernstein's conjecture implies the corresponding entropy inequality at the exponent $r = 2$. Lei and Zhang recently proved this inequality and determined its equality cases [LZ26, Theorem 1.3].

1.3. **Sketch of the proof and organization of the paper.** The proof of Theorem 1.1 consists of four main steps.

**Step 1: Polar factorization and entropy reduction.** Section 2 begins with a self-inversive normalization. Under the simple-zero assumption, we define

$$q = p - \frac{1}{n} zp', \qquad B = \frac{q^*}{q},$$

where the reflection is taken relative to the fixed degree $n$, and prove that $q$ is zero-free on $\overline{\mathbb{D}}$, that $B$ is a finite Blaschke product satisfying $B(0) = 0$, and that

$$p = q + q^* = q(1 + B).$$

Moreover, $B$ preserves normalized Haar measure on $\mathbb{T}$.

Differentiating

$$r \longmapsto \log \frac{\|p\|_r}{\|Q_n\|_r}$$

with respect to $r$ expresses its derivative as a difference of relative entropies. An entropy decomposition then reduces the desired monotonicity to the centered polar inequality

$$\int_{\mathbb{T}} |q|^r |1 + B|^r \big(\log |1 + B| - \ell_r\big)\ \mathrm{d}m \geq 0, \tag{1.2}$$

where $\ell_r := \dfrac{\displaystyle\int_{\mathbb{T}} |1 + \zeta|^r \log |1 + \zeta|\ \mathrm{d}m(\zeta)}{\displaystyle\int_{\mathbb{T}} |1 + \zeta|^r\ \mathrm{d}m(\zeta)}$.

**Step 2: Fourier and area reduction.** In Section 3, we write $r = 2a$ and choose the analytic branch $g = q^a$. A Fourier expansion, an Euler-operator recurrence, and a polarized Dirichlet identity yield the exact representation

$$\begin{aligned}\int_{\mathbb{T}} |q|^{2a} |1 + B|^{2a} \big(\log |1 + B| - \ell_{2a}\big)\ \mathrm{d}m \\ = \frac{M_{Q_n}(2a)}{an} \int_{\mathbb{D}} \operatorname{Re} H_a(B(z))\, |(q^a)'(z)|^2\ \mathrm{d}A(z),\end{aligned} \tag{1.3}$$

where $M_{Q_n}(2a) := \int_{\mathbb{T}} |1 + \zeta|^{2a}\ \mathrm{d}m(\zeta)$, $\mathrm{d}A(z) := \mathrm{d}x\ \mathrm{d}y/\pi$, with $z = x + iy$, denotes normalized area measure on $\mathbb{D}$, and $H_a$ is a universal analytic kernel independent of the degree and the zero configuration. Thus it remains to prove

$$\operatorname{Re} H_a(z) > 0, \qquad a > 0, \quad z \in \mathbb{D}. \tag{1.4}$$

**Step 3: Positivity of the universal kernel.** The kernel inequality (1.4) is proved separately in two complementary parameter ranges. Section 4 treats $0 < a \le 1$ by establishing a sharp summable estimate for the Taylor coefficients of $H_a$. Section 5 treats $a \ge 1$ by means of a Joukowski–Stieltjes representation, a beta transform, a positive Abel decomposition, and finite-basis inequalities. Together, these two arguments establish (1.4) for every $a > 0$.

**Step 4: Completion of the proof.** By (1.3), the kernel positivity (1.4) implies the centered polar inequality (1.2). This yields the relative-entropy comparison and hence

$$\frac{\mathrm{d}}{\mathrm{d}r} \log \frac{\|p\|_r}{\|Q_n\|_r} \ge 0, \qquad r > 0,$$

for polynomials with simple unimodular zeros.

Section 6 removes the simple-zero assumption by approximation and treats the endpoint exponents $r = 0$ and $r = \infty$. Integrating the preceding differential inequality then gives (1.1), thereby completing the proof of Theorem 1.1.

The organization of the paper follows the preceding four steps. Section 2 develops the polar factorization and entropy reduction. Section 3 derives the Fourier and area identities. Sections 4 and 5 establish kernel positivity in the ranges $0 < a \le 1$ and $a \ge 1$, respectively. Section 6 completes the proof of the theorem, and Section 7 presents its principal applications and sharp consequences.

The logical dependencies among the principal results are summarized in Figure 1.

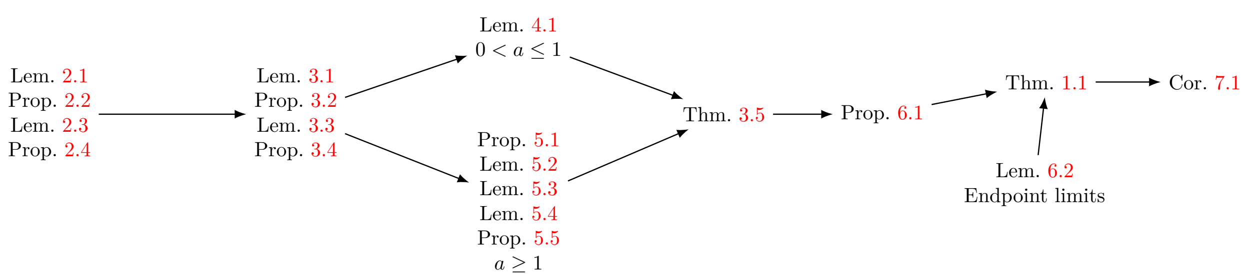


FIGURE 1. Logical dependencies among the principal results.

## 2. NORMALIZATION AND ENTROPY REDUCTION

This section isolates the two structural ingredients used throughout the proof: a polar factorization for self-inversive polynomials and the entropy identity obtained by differentiating an $L^r$ mean. The objects introduced here recur in the sequel; the auxiliary notation used in later sections will be defined locally.

Fix an integer $n \ge 1$. For every polynomial $f$ of degree at most $n$, its reflection relative to the fixed degree $n$ is

$$f^*(z) := z^n \overline{f(1/\bar{z})}. \tag{2.1}$$

We also write

$$\mathcal{D}f(z) := zf'(z) \tag{2.2}$$

for the Euler operator.

2.1. **Self-inversive normalization and the polar factor.** We begin by removing the inessential scalar and rotational degrees of freedom.

**Lemma 2.1.** *Let $P$ be a nonzero polynomial of degree $n$ whose zeros lie on $\mathbb{T}$. Then there are $c \neq 0$ and $\eta \in \mathbb{T}$ such that*

$$p(z) := cP(\eta z) = 1 + a_1 z + \cdots + a_{n-1} z^{n-1} + z^n \quad \textit{and} \quad p = p^*, \tag{2.3}$$

*where $p^*$ is defined by (2.1). Moreover,*

$$\|P\|_r = |c|^{-1} \|p\|_r, \qquad 0 \le r \le \infty. \tag{2.4}$$

*Finally, for $Q_n(z) = 1 + z^n$,*

$$\|Q_n\|_r = 2\left(\frac{\Gamma((r+1)/2)}{\sqrt{\pi}\,\Gamma(1+r/2)}\right)^{1/r}, \qquad 0 < r < \infty, \tag{2.5}$$

*while $\|Q_n\|_0 = 1$ and $\|Q_n\|_\infty = 2$.*

*Proof.* Write

$$P(z) = c_n \prod_{j=1}^{n} (z - \zeta_j), \qquad |\zeta_j| = 1,$$

and let $c_0 = P(0)$. Then

$$\alpha := \frac{c_0}{c_n} = (-1)^n \prod_{j=1}^{n} \zeta_j \in \mathbb{T}.$$

Choose $\eta \in \mathbb{T}$ with $\eta^n = \alpha$ and set $c = c_0^{-1}$. The polynomial $p(z) = cP(\eta z)$ has constant and leading coefficients equal to 1. Its zeros remain on $\mathbb{T}$, so $p$ and $p^*$ are monic polynomials with the same zeros; hence $p = p^*$. Equation (2.4) follows from rotation invariance of the normalized Haar measure $m$.

The map $z \mapsto z^n$ preserves the normalized Haar measure $m$, and therefore $\|Q_n\|_r = \|1+z\|_r$. The substitution $z = \mathrm{e}^{i\theta}$ followed by the beta integral gives (2.5). Jensen's formula gives the limit at $r = 0$, while the supremum of $|1 + z^n|$ on $\mathbb{T}$ is 2. □

For the rest of Sections 2–5, $p$ denotes a polynomial satisfying (2.3). Until Section 6, we also assume that the zeros of $p$ are simple. The following polar factorization is the point at which this hypothesis enters.

**Proposition 2.2.** *Let $p$ satisfy (2.3) and have simple zeros on $\mathbb{T}$. Define*

$$q := p - \frac{1}{n}\mathcal{D}p, \qquad B := \frac{q^*}{q}, \tag{2.6}$$

*where $\mathcal{D}$ and the reflection are given by (2.2) and (2.1). Then $q$ has no zeros on $\overline{\mathbb{D}}$, $B$ is a finite Blaschke product with $B(0) = 0$, and*

$$q^* = \frac{1}{n}\mathcal{D}p, \qquad p = q + q^* = q(1 + B). \tag{2.7}$$

*Moreover, $B$ preserves normalized Haar measure:*

$$\int_{\mathbb{T}} \varphi(B(z))\, \mathrm{d}m(z) = \int_{\mathbb{T}} \varphi(\zeta)\, \mathrm{d}m(\zeta) \tag{2.8}$$

*for every $\varphi \in L^1(\mathbb{T}, m)$.*

*Proof.* The identities in (2.7) follow by comparing coefficients; see also [LZ26, p. 5, (2.4)–(2.5)], while the zero-free property of $q$ and the fact that $B = q^*/q$ is a finite Blaschke product with $B(0) = 0$ follow from [LZ26, p. 5, Lemma 2.2].

It remains to prove (2.8). Since $B(0) = 0$, the mean-value property gives

$$\int_{\mathbb{T}} B^k \, \mathrm{d}m = 0, \qquad k \ge 1.$$

Because $|B| = 1$ on $\mathbb{T}$, the negative Fourier coefficients vanish as well. Hence the finite Borel measures $B_* m$ and $m$ have the same Fourier coefficients. Uniqueness of Fourier coefficients for finite measures gives $B_* m = m$, and therefore (2.8) holds for every integrable $\varphi$. This observation also appears in [Nor68, p. 442, Lemma 1 and its corollary]. □

2.2. **Differentiation in the exponent.** For a nonzero polynomial $f$ and $r > 0$, define its $r$-moment and the associated probability measure by

$$M_f(r) := \int_{\mathbb{T}} |f|^r \, \mathrm{d}m, \qquad \mathrm{d}\rho_{f,r} := \frac{|f|^r}{M_f(r)} \, \mathrm{d}m. \tag{2.9}$$

For a probability measure $\rho \ll m$, its relative entropy with respect to $m$ is

$$\mathsf{D}(\rho \| m) := \int_{\mathbb{T}} \log \left( \frac{\mathrm{d}\rho}{\mathrm{d}m} \right) \, \mathrm{d}\rho, \tag{2.10}$$

with the convention $0 \log 0 = 0$. The next identity shows that differentiation of an $L^r$-mean produces this entropy.

**Lemma 2.3.** *Let $f$ be a nonzero polynomial, and let $M_f(r)$ and $\rho_{f,r}$ be defined by (2.9). Then, for every $r > 0$,*

$$r^2 \frac{\mathrm{d}}{\mathrm{d}r} \log \|f\|_r = r \frac{M_f'(r)}{M_f(r)} - \log M_f(r) = \mathsf{D}(\rho_{f,r} \| m). \tag{2.11}$$

*In particular, let $p$ be the normalized polynomial in (2.3), and set $Q_n(z) := 1 + z^n$. If*

$$\mathsf{D}(\rho_{p,r} \| m) \geq \mathsf{D}(\rho_{Q_n,r} \| m) \tag{2.12}$$

*holds for every $r > 0$, then $r \mapsto \log(\|p\|_r \,/\, \|Q_n\|_r)$ is nondecreasing on $(0, \infty)$.*

*Proof.* Fix $r_0 > 0$. Near a zero $\mathrm{e}^{i\theta_0}$ of multiplicity $m_0$, one has $|f(\mathrm{e}^{i\theta})| \asymp |\theta - \theta_0|^{m_0}$. Hence, uniformly for $u \in [r_0/2, 3r_0/2]$,

$$|f(\mathrm{e}^{i\theta})|^u |\log |f(\mathrm{e}^{i\theta})|| \leq C |\theta - \theta_0|^{m_0 r_0/2} \big(1 + |\log |\theta - \theta_0||\big),$$

which is integrable. Away from the finitely many zeros the integrand is uniformly bounded. Differentiation under the integral sign is therefore justified and gives

$$M_f'(r) = \int_{\mathbb{T}} |f|^r \log |f| \, \mathrm{d}m.$$

Since $\log \|f\|_r = r^{-1} \log M_f(r)$, the first identity in (2.11) follows by differentiation. Substituting the density in (2.9) into (2.10) gives the second identity. Applying the identity to $p$ and $Q_n$ gives

$$\frac{\mathrm{d}}{\mathrm{d}r} \log \frac{\|p\|_r}{\|Q_n\|_r} = \frac{1}{r^2} \left[ \mathsf{D}(\rho_{p,r} \| m) - \mathsf{D}(\rho_{Q_n,r} \| m) \right],$$

which proves the final assertion. □

2.3. **The centered polar term.** Let $p, q, B$ be as in Proposition 2.2. For $r > 0$, define

$$\mathrm{d}\mu_r := \frac{|1 + B|^r}{M_{Q_n}(r)} \, \mathrm{d}m, \qquad \ell_r := \int_{\mathbb{T}} \log |1 + B| \, \mathrm{d}\mu_r. \tag{2.13}$$

Proposition 2.2 implies that $\mu_r$ is a probability measure and that $\ell_r$ depends only on $r$, not on $p$. We also introduce the centered polar functional

$$\mathcal{P}_r(q, B) := \int_{\mathbb{T}} |q|^r |1 + B|^r \big( \log |1 + B| - \ell_r \big) \, \mathrm{d}m. \tag{2.14}$$

For a probability measure $\mu$ and a nonnegative integrable function $w$, define

$$\mathrm{Ent}_\mu(w) := \int w \log w \, \mathrm{d}\mu - \left( \int w \, \mathrm{d}\mu \right) \log \left( \int w \, \mathrm{d}\mu \right). \tag{2.15}$$

The next decomposition separates an automatically nonnegative entropy term from the only term whose sign remains to be proved.

**Proposition 2.4.** *Let $p, q, B$ be as in Proposition 2.2. For $r > 0$, let $M_f(r)$ and $\rho_{f,r}$ be as in (2.9), let $\mu_r$, $\ell_r$, and $\mathcal{P}_r(q, B)$ be defined by (2.13) and (2.14), and let $\mathrm{Ent}_\mu$ be defined by (2.15). Then*

$$\mathsf{D}(\rho_{p,r}\|m) - \mathsf{D}(\rho_{Q_n,r}\|m) = \frac{M_{Q_n}(r)}{M_p(r)}\,\mathrm{Ent}_{\mu_r}(|q|^r) + \frac{r}{M_p(r)}\,\mathcal{P}_r(q, B). \tag{2.16}$$

*In particular, the entropy comparison (2.12) follows once*

$$\mathcal{P}_r(q, B) \geq 0 \tag{2.17}$$

*has been proved.*

*Proof.* Set $w := |q|^r$ and $\Lambda := \int_{\mathbb{T}} w \,\mathrm{d}\mu_r = \frac{M_p(r)}{M_{Q_n}(r)}$. By (2.7),

$$\mathrm{d}\rho_{p,r} = \frac{w}{\Lambda}\,\mathrm{d}\mu_r.$$

Define the probability measure

$$\mathrm{d}\nu_r(\zeta) := \frac{|1+\zeta|^r}{M_{Q_n}(r)}\,\mathrm{d}m(\zeta).$$

By (2.8), one has

$$B_*\mu_r = \nu_r,$$

where $B_*\mu_r$ denotes the pushforward of $\mu_r$ under $B$. Indeed, for every bounded Borel function $\varphi$,

$$\begin{aligned}
\int_{\mathbb{T}} \varphi \,\mathrm{d}(B_*\mu_r) &= \int_{\mathbb{T}} \varphi(B)\,\mathrm{d}\mu_r \\
&= \frac{1}{M_{Q_n}(r)} \int_{\mathbb{T}} \varphi(B)|1+B|^r\,\mathrm{d}m \\
&= \frac{1}{M_{Q_n}(r)} \int_{\mathbb{T}} \varphi(\zeta)|1+\zeta|^r\,\mathrm{d}m(\zeta) = \int_{\mathbb{T}} \varphi\,\mathrm{d}\nu_r,
\end{aligned}$$

where the third equality follows from $B_*m = m$. Hence $B_*\mu_r = \nu_r$.

Let $\pi_n : \mathbb{T} \to \mathbb{T}$, $\pi_n(z) := z^n$. Since $(\pi_n)_*m = m$ and $\frac{\mathrm{d}\rho_{Q_n,r}}{\mathrm{d}m} = \left(\frac{\mathrm{d}\nu_r}{\mathrm{d}m}\right) \circ \pi_n$, we also have

$$(\pi_n)_*\rho_{Q_n,r} = \nu_r.$$

Consequently,

$$\begin{aligned}
\mathsf{D}(\rho_{Q_n,r}\|m) &= \int_{\mathbb{T}} \log\left(\frac{\mathrm{d}\nu_r}{\mathrm{d}m} \circ \pi_n\right)\left(\frac{\mathrm{d}\nu_r}{\mathrm{d}m} \circ \pi_n\right)\,\mathrm{d}m \\
&= \int_{\mathbb{T}} \log\left(\frac{\mathrm{d}\nu_r}{\mathrm{d}m}\right)\frac{\mathrm{d}\nu_r}{\mathrm{d}m}\,\mathrm{d}m \\
&= \mathsf{D}(\nu_r\|m).
\end{aligned}$$

All logarithmic terms below are integrable, since near each zero they are controlled by a constant multiple of $t^r(1 + |\log t|)$ with $r > 0$; hence the pushforward identities extend from bounded functions by truncation. Thus,

$$\mathsf{D}(\rho_{Q_n,r}\|m) = \int_{\mathbb{T}} \log\left(\frac{|1+B|^r}{M_{Q_n}(r)}\right)\,\mathrm{d}\mu_r.$$

Subtracting the two entropy formulas yields

$$\mathsf{D}(\rho_{p,r}\|m) - \mathsf{D}(\rho_{Q_n,r}\|m) = \frac{1}{\Lambda}\int_{\mathbb{T}} w \log\frac{w}{\Lambda}\,\mathrm{d}\mu_r + \int_{\mathbb{T}} \left(\frac{w}{\Lambda} - 1\right)\log\left(\frac{|1+B|^r}{M_{Q_n}(r)}\right)\,\mathrm{d}\mu_r.$$

The first term equals $\Lambda^{-1}\,\mathrm{Ent}_{\mu_r}(w)$. In the second, the constant $-\log M_{Q_n}(r)$ disappears because $\int (w/\Lambda - 1)\,\mathrm{d}\mu_r = 0$. Centering the remaining logarithm by $\ell_r$ gives

$$\frac{r}{\Lambda}\int_{\mathbb{T}} w\big(\log|1+B| - \ell_r\big)\,\mathrm{d}\mu_r = \frac{r}{M_p(r)}\mathcal{P}_r(q, B).$$

This proves (2.16). Finally, $\mathrm{Ent}_{\mu_r}(|q|^r) \geq 0$ by Jensen's inequality for $x \mapsto x \log x$, so (2.17) implies (2.12). $\square$

The remainder of the proof is therefore devoted to (2.17).

## 3. A boundary-to-area identity

Fix $a > 0$ and write $r = 2a$. Let $p$ satisfy the hypotheses of Proposition 2.2, and let $q$ and $B$ be defined by (2.6). Since $q$ is zero-free on a neighborhood of $\overline{\mathbb{D}}$ and $q(0) = 1$, there is a unique analytic logarithm $L$ of $q$ with $L(0) = 0$. Throughout this section,

$$g := q^a := \exp(aL), \tag{3.1}$$

so that $|g|^2 = |q|^{2a}$ on $\mathbb{T}$. We use normalized area measure

$$\mathrm{d}A(z) := \frac{\mathrm{d}x\ \mathrm{d}y}{\pi}, \qquad z = x + iy. \tag{3.2}$$

The aim is to represent the centered polar functional $\mathcal{P}_{2a}(q, B)$ from (2.14) as a weighted Dirichlet integral.

### 3.1. The model Fourier coefficients.

Since $z \mapsto z^n$ preserves the normalized Haar measure $m$, $M_{Q_n}(2a) = \int_{\mathbb{T}} |1 + \zeta|^{2a}\ \mathrm{d}m(\zeta)$. Define the probability density

$$h_a(\zeta) := \frac{|1 + \zeta|^{2a}}{M_{Q_n}(2a)}, \qquad \zeta \in \mathbb{T}, \tag{3.3}$$

and its Fourier coefficients

$$\alpha_k(a) := \int_{\mathbb{T}} h_a(\zeta)\zeta^{-k}\ \mathrm{d}m(\zeta), \qquad k \in \mathbb{Z}. \tag{3.4}$$

A dot will always mean differentiation with respect to $a$; thus $\dot\alpha_k(a) = \partial_a \alpha_k(a)$. The relevant Fourier coefficients are explicit and have a summable derivative.

**Lemma 3.1.** *Let $a > 0$, and let $h_a$ and $\alpha_k(a)$ be defined by (3.3) and (3.4). Then*

$$\alpha_0(a) = 1, \qquad \alpha_{-k}(a) = \alpha_k(a) \in \mathbb{R}.$$

*For every integer $k \geq 1$,*

$$\alpha_k(a) = \frac{\Gamma(a+1)^2}{\Gamma(a-k+1)\Gamma(a+k+1)} = \prod_{j=0}^{k-1} \frac{a-j}{a+j+1}. \tag{3.5}$$

*At integer values of $a$, the gamma quotient is interpreted through the entire reciprocal gamma function; equivalently, one may use the product formula in (3.5). Moreover,*

$$\dot\alpha_k(a) = O_a\big(k^{-2a-1}\log(k+1)\big) \qquad (k \to \infty). \tag{3.6}$$

*This estimate is locally uniform in $a$: if $K \Subset (0, \infty)$ and $a_- := \min K$, then*

$$\sup_{a \in K} |\dot\alpha_k(a)| \leq C_K k^{-2a_- - 1}\log(k+1), \qquad k \geq 1.$$

*If $a = N \in \mathbb{N}$ and $k \geq N + 1$, then*

$$\dot\alpha_k(N) = (-1)^{k-N-1}\frac{(N!)^2(k-N-1)!}{(k+N)!}. \tag{3.7}$$

*In particular,*

$$\sum_{k \geq 1} |\dot\alpha_k(a)| < \infty. \tag{3.8}$$

*Proof.* The symmetry assertions follow because $h_a(\mathrm{e}^{i\theta})$ is real and even. With $\theta = 2t$,

$$\alpha_k(a) = \frac{\displaystyle\int_0^{\pi/2} \cos^{2a} t \cos(2kt)\ \mathrm{d}t}{\displaystyle\int_0^{\pi/2} \cos^{2a} t\ \mathrm{d}t}.$$

The cosine–beta integral [GR14, Formula 3.631.9] gives the gamma quotient in (3.5); the product form follows from $\Gamma(z+1) = z\Gamma(z)$.

Euler's reflection formula rewrites (3.5), by analytic continuation in $a$, as

$$\alpha_k(a) = (-1)^{k+1} c(a) \frac{\Gamma(k-a)}{\Gamma(k+a+1)}, \qquad c(a) := \frac{\sin \pi a}{\pi} \Gamma(a+1)^2,$$

whenever $k > a$. Writing $\psi := \Gamma'/\Gamma$ for the digamma function and differentiating this identity gives

$$\dot\alpha_k(a) = (-1)^{k+1} \frac{\Gamma(k-a)}{\Gamma(k+a+1)} \left\{ c'(a) - c(a)\big[\psi(k-a) + \psi(k+a+1)\big] \right\}.$$

Let $K \Subset (0,\infty)$ and $a_- := \min K$. For all sufficiently large $k$, uniformly for $a \in K$, the gamma-ratio expansion [OLBC10, p. 141, Eq. (5.11.13)] gives

$$\frac{\Gamma(k-a)}{\Gamma(k+a+1)} = O_K\big(k^{-2a-1}\big).$$

Since $a \ge a_-$ on $K$, this is also $O_K(k^{-2a_- -1})$. The standard asymptotic formula for the digamma function [OLBC10, p. 140, Eq. (5.11.2)] gives

$$\psi(k-a) + \psi(k+a+1) = O_K(\log k).$$

Since $c$ and $c'$ are bounded on $K$, these estimates prove (3.6) and its stated local uniformity. If $a = N \in \mathbb{N}$ and $k > N$, the product in (3.5) has a single vanishing factor, namely $a - N$; differentiating that factor gives (3.7). These estimates imply (3.8). □

3.2. **Fourier reduction.** For $k \ge 0$, define the boundary correlations

$$c_k := \int_{\mathbb{T}} B^k |g|^2\ \mathrm{d}m = \langle B^k g, g\rangle_{H^2}, \tag{3.9}$$

where the $H^2$ inner product is linear in the first variable. Differentiating the model density converts the polar functional into these correlations.

**Proposition 3.2.** *Fix $a > 0$. Let $p, q, B$ be as in Proposition 2.2, let $g = q^a$ be defined by (3.1), and let $c_k$ be defined by (3.9). With $\dot\alpha_k(a)$ as in Lemma 3.1, one has*

$$\mathcal{P}_{2a}(q,B) = M_{Q_n}(2a) \sum_{k\ge1} \dot\alpha_k(a)\, \mathrm{Re}\, c_k. \tag{3.10}$$

*Proof.* Differentiating (3.3) in $a$ gives

$$\dot h_a(\zeta) = 2h_a(\zeta)\big(\log|1+\zeta| - \ell_{2a}\big).$$

Fix a compact interval $K \Subset (0,\infty)$ containing $a$, and put $a_- := \min K$. Since $M_{Q_n}(2u)$ is positive and $C^1$ for $u \in K$, there is a constant $C_K$ such that, uniformly for $u \in K$,

$$|\partial_u h_u(\zeta)| \le C_K |1+\zeta|^{2a_-}\big(1 + |\log|1+\zeta||\big)$$

near $\zeta = -1$. The right-hand side is integrable, while away from $-1$ the derivatives are uniformly bounded. Hence the Fourier coefficients may be differentiated under the integral sign. Moreover, $t^{2a}|\log t| \to 0$ as $t \downarrow 0$, so $\dot h_a$ extends continuously to $\zeta = -1$. Since $\alpha_0(a) = 1$, the zeroth Fourier coefficient may be differentiated under the integral sign. Since $\alpha_0(a) = 1$, the zeroth Fourier coefficient of $\dot{h}_a$ is zero. By (3.8), the series

$$\sum_{k\geq 1} \dot{\alpha}_k(a)\big(\zeta^k + \zeta^{-k}\big)$$

converges absolutely and uniformly on $\mathbb{T}$. Its Fourier coefficients agree at every integer index with those of the continuous function $\dot{h}_a$. Uniqueness of Fourier coefficients in $L^1(\mathbb{T})$ therefore gives equality almost everywhere; since both sides are continuous, the equality holds everywhere. Thus

$$\dot{h}_a(\zeta) = \sum_{k\geq 1} \dot{\alpha}_k(a)\big(\zeta^k + \zeta^{-k}\big).$$

Together with

$$\dot{h}_a(\zeta) = 2h_a(\zeta)\big(\log|1+\zeta| - \ell_{2a}\big),$$

this yields

$$h_a(\zeta)\big(\log|1+\zeta| - \ell_{2a}\big) = \frac{1}{2}\sum_{k\geq 1} \dot{\alpha}_k(a)\big(\zeta^k + \zeta^{-k}\big).$$

Substitute $\zeta = B$, multiply by $M_{Q_n}(2a)|g|^2$, and integrate over $\mathbb{T}$. Since $|B| = 1$ on $\mathbb{T}$,

$$\int_{\mathbb{T}} B^{-k}|g|^2 \, \mathrm{d}m = \overline{c_k}.$$

Equation (3.10) follows. □

3.3. **The Dirichlet recurrence.** For $k \geq 0$, define the area moments

$$V_k := \int_{\mathbb{D}} B^k |g'|^2 \, \mathrm{d}A. \tag{3.11}$$

We shall use the polarized Dirichlet identity

$$\langle \mathcal{D}u, v\rangle_{H^2} = \int_{\mathbb{D}} u'\overline{v'} \, \mathrm{d}A, \qquad u(0) = 0, \tag{3.12}$$

valid for functions analytic on a neighborhood of $\overline{\mathbb{D}}$. It follows by comparing Taylor coefficients; see also [EKMR14, pp. 1–2, Section 1.1]. The boundary and area moments are linked by the following recurrence.

**Lemma 3.3.** *Fix $a > 0$. Let $p, q, B$ be as in Proposition 2.2, let $g = q^a$ be defined by (3.1), and let $c_k$ and $V_k$ be defined by (3.9) and (3.11). Then, for every integer $k \geq 1$,*

$$akn\, c_k = (a+k)V_{k-1} - (a-k)V_k. \tag{3.13}$$

*Proof.* Set

$$\sigma := \frac{\mathcal{D}q}{q}, \qquad I_k := \int_{\mathbb{T}} B^k \sigma |g|^2 \, \mathrm{d}m \quad (k \geq 0).$$

From $p = q(1+B)$ and $\mathcal{D}p = nqB$, we obtain

$$(1+B)\sigma = nB - \mathcal{D}B. \tag{3.14}$$

On $\mathbb{T}$, the identity $B = q^*/q$ gives

$$\overline{\sigma} = \frac{\sigma}{B}. \tag{3.15}$$

Indeed,

$$\frac{\mathcal{D}B}{B} = n - \sigma - \overline{\sigma}$$

by differentiating $B = q^*/q$ on $\mathbb{T}$, whereas (3.14) gives $\mathcal{D}B/B = n - \sigma - \sigma/B$.

For $k \geq 1$, set

$$X_k := \int_{\mathbb{T}} B^{k-1} \mathcal{D}B \, |g|^2 \, \mathrm{d}m.$$

Multiplying (3.14) by $B^{k-1}|g|^2$ gives

$$I_{k-1} + I_k = nc_k - X_k.$$

Since $\mathcal{D}g = a\sigma g$ and the Euler operator is symmetric for the $H^2$ inner product on these analytic functions,

$$kX_k + aI_k = \langle \mathcal{D}(B^k g), g \rangle_{H^2} = \langle B^k g, \mathcal{D}g \rangle_{H^2} = aI_{k-1},$$

where the last equality uses (3.15). Eliminating $X_k$ yields

$$knc_k + (a-k)I_k - (a+k)I_{k-1} = 0. \tag{3.16}$$

Finally, choose an analytic primitive $u_k$ with $u_k' = B^k g'$ and $u_k(0) = 0$. Then (3.12) and $\mathcal{D}g = a\sigma g$ give

$$V_k = \int_{\mathbb{D}} B^k |g'|^2 \, \mathrm{d}A = \langle \mathcal{D}u_k, g \rangle_{H^2} = \langle B^k \mathcal{D}g, g \rangle_{H^2} = aI_k.$$

Substitution into (3.16) proves (3.13). □

3.4. **The universal kernel.** Define, for $a > 0$ and $z \in \mathbb{D}$,

$$H_a(z) := \sum_{k \geq 1} \frac{\dot{\alpha}_k(a)}{k} \Big( (a+k)z^{k-1} - (a-k)z^k \Big). \tag{3.17}$$

Lemma 3.1 implies absolute and uniform convergence on $\overline{\mathbb{D}}$. Thus $H_a$ is analytic on $\mathbb{D}$ and continuous on $\overline{\mathbb{D}}$. Combining the Fourier reduction with the Dirichlet recurrence gives the desired Hardy–Bergman identity.

**Proposition 3.4.** *Let $a > 0$, let $p, q, B$ be as in Proposition 2.2, let $\mathcal{P}_{2a}(q, B)$ be defined by (2.14), let $g = q^a$ and $\mathrm{d}A$ be defined by (3.1) and (3.2), and let $H_a$ be defined by (3.17). Then*

$$\mathcal{P}_{2a}(q, B) = \frac{M_{Q_n}(2a)}{an} \int_{\mathbb{D}} \operatorname{Re} H_a(B(z)) \, |g'(z)|^2 \, \mathrm{d}A(z). \tag{3.18}$$

*Proof.* Since $|B| \leq 1$ in $\mathbb{D}$, the moments in (3.11) satisfy $|V_k| \leq V_0$. Moreover,

$$\sum_{k \geq 1} \frac{|\dot{\alpha}_k(a)|}{k} \big( (a+k) + |a-k| \big) \leq 2(1+a) \sum_{k \geq 1} |\dot{\alpha}_k(a)| < \infty.$$

Hence Lemma 3.1 permits termwise summation. Multiplying (3.13) by $\dot{\alpha}_k(a)/(akn)$, summing over $k \geq 1$, and taking real parts gives

$$\sum_{k \geq 1} \dot{\alpha}_k(a) \operatorname{Re} c_k = \frac{1}{an} \int_{\mathbb{D}} \operatorname{Re} H_a(B(z)) \, |g'(z)|^2 \, \mathrm{d}A(z).$$

Now apply Proposition 3.2. □

The proof of Theorem 1.1 has now been reduced to the following universal positivity statement, which no longer involves $p$, $q$, $B$, or the degree $n$.

**Theorem 3.5.** *Let $H_a$ be the analytic function defined by (3.17). Then*

$$\operatorname{Re} H_a(z) > 0, \qquad a > 0, \quad z \in \mathbb{D}. \tag{3.19}$$

Indeed, Theorem 3.5 and (3.18) imply $\mathcal{P}_{2a}(q, B) \geq 0$. Proposition 2.4 then gives the entropy comparison (2.12). Sections 4 and 5 prove Theorem 3.5 in the ranges $0 < a \leq 1$ and $a \geq 1$, respectively.

## 4. Kernel positivity for $0 < a \le 1$

Let $H_a$ be the kernel in (3.17). Write

$$H_a(z) = \beta_0(a) + \sum_{k\ge1} \beta_k(a) z^k. \tag{4.1}$$

Collecting adjacent terms in (3.17) gives

$$\beta_0(a) = \frac{1}{a+1}, \qquad \beta_k(a) = \frac{a+k+1}{k+1}\dot{\alpha}_{k+1}(a) - \frac{a-k}{k}\dot{\alpha}_k(a) \quad (k \ge 1), \tag{4.2}$$

where $\alpha_k(a)$ is defined by (3.4). The small-parameter argument rests on the following sharp summable estimate.

**Lemma 4.1.** *Let $0 < a \le 1$, and let $\beta_k(a)$ be defined by (4.1)–(4.2). Then*

$$|\beta_k(a)| \le \frac{1}{(a+1)k(k+1)}, \qquad k \ge 1. \tag{4.3}$$

*Proof.* Assume first that $0 < a < 1$. For $k \ge 1$, set

$$u_k := (-1)^{k-1}\alpha_k(a) = \frac{a(1-a)_{k-1}}{(a+1)_k} > 0, \qquad C_k := (-1)^{k-1}k(k+1)\beta_k(a),$$

where $(x)_j := \Gamma(x+j)/\Gamma(x)$ is the Pochhammer symbol, and a prime denotes differentiation with respect to $a$. Differentiating

$$(a+k+1)\alpha_{k+1}(a) = (a-k)\alpha_k(a)$$

and using (4.2), we obtain

$$C_k = \frac{k(2k+1)}{a+k+1}u_k + (k-a)u_k'. \tag{4.4}$$

Set

$$R_k := \prod_{j=1}^{k} \frac{j-a}{j+a}, \qquad S_k := \sum_{j=1}^{k} \frac{2j}{j^2-a^2}.$$

Since $(k-a)u_k = aR_k$ and $R_k'/R_k = -S_k$, equation (4.4) becomes

$$C_k = R_k\big(1 - aS_k + T_k\big), \qquad T_k := 2a + \frac{a(a+1)(2a+1)}{(k-a)(k+a+1)}. \tag{4.5}$$

We first prove the lower bound. Put $h_1 := 0, h_k := \sum_{j=2}^{k} \frac{1}{j}$ and $x := 2ah_k$ The inequality $\log((1-u)/(1+u)) \le -2u$ for $0 < u < 1$ gives

$$R_k \le \frac{1-a}{1+a}\mathrm{e}^{-x}.$$

Also,

$$S_k \le \frac{2}{1-a^2} + \frac{2h_k}{1-a^2/4}.$$

Consequently,

$$(a+1)aR_kS_k \le \mathrm{e}^{-x}\left[\frac{2a}{1+a} + \frac{1-a}{1-a^2/4}x\right] \le \mathrm{e}^{-x}(1+x) \le 1,$$

where the middle inequality uses $2a/(1+a) \le 1$ and $(1-a)/(1-a^2/4) \le 1$. Since $R_k(1+T_k) > 0$, (4.5) yields

$$C_k \ge -aR_kS_k \ge -\frac{1}{a+1}. \tag{4.6}$$

For the upper bound, use $S_k \geq 2/(1-a^2)$ in (4.5):

$$C_k \leq R_k U_k, \qquad U_k := 1 + T_k - \frac{2a}{1-a^2}.$$

The sequences $R_k$ and $T_k$ are decreasing; hence so is $U_k$. If $U_k \leq 0$, then $C_k \leq 0$. If $k \geq 2$ and $U_k > 0$, then $0 < U_k \leq U_2$ and $0 < R_k \leq R_2$. Hence

$$C_k \leq R_k U_k \leq R_2 U_2 < \frac{1}{a+1},$$

because

$$1 - (a+1)R_2U_2 = \frac{a(11 + 10a + 11a^2)}{(a+1)(a+2)(a+3)} > 0.$$

For $k = 1$, direct substitution gives

$$C_1 = \frac{2(a^2 + a + 1)}{(a+1)^2(a+2)} \leq \frac{1}{a+1},$$

the last inequality being equivalent to $a^2 \leq a$. Together with (4.6), this proves $|C_k| \leq 1/(a+1)$, which proves (4.3).

Finally, each $\beta_k(a)$ is continuous for $a > 0$, so the estimate extends to $a = 1$. Equivalently, one may check directly that

$$\beta_0(1) = \frac{1}{2}, \qquad \beta_1(1) = \frac{1}{4}, \qquad \beta_k(1) = \frac{(-1)^k}{k^2(k+1)^2} \quad (k \geq 2).$$

□

We now conclude the proof of Theorem 3.5 in the range $0 < a \leq 1$.

*Proof of Theorem 3.5 for* $0 < a \leq 1$. Let $|z| = \rho < 1$. By Lemma 4.1,

$$\operatorname{Re} H_a(z) \geq \frac{1}{a+1}\left(1 - \sum_{k\geq 1} \frac{\rho^k}{k(k+1)}\right) > 0,$$

because $\sum_{k\geq 1} 1/[k(k+1)] = 1$ and $\rho^k < 1$. This proves (3.19) in the stated range. □

## 5. Kernel positivity for $a \geq 1$

Throughout this section, $a \geq 1$ is fixed. Let

$$\mathrm{B}(x,y) := \int_0^1 t^{x-1}(1-t)^{y-1}\, \mathrm{d}t = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}, \qquad x, y > 0,$$

denote the Euler beta function. Set

$$Z_a := \mathrm{B}\left(a + \frac{1}{2}, \frac{1}{2}\right) = \frac{\sqrt{\pi}\,\Gamma(a+1/2)}{\Gamma(a+1)}, \qquad \omega_a(x) := \frac{x^{a-1/2}(1-x)^{-1/2}}{Z_a}, \quad 0 < x < 1. \tag{5.1}$$

We use the Pochhammer symbol $(x)_m := \Gamma(x+m)/\Gamma(x)$ for $m \geq 0$. Define also

$$m_a := \int_0^1 \omega_a(x) \log x\, \mathrm{d}x = \psi\left(a + \frac{1}{2}\right) - \psi(a+1) < 0. \tag{5.2}$$

Thus $\omega_a$ is a probability density. Define its centered primitive and the associated first-order expression by

$$W_a(x) := -\int_0^x \omega_a(u)(\log u - m_a)\, \mathrm{d}u, \qquad R_a(x) := aW_a(x) - xW_a'(x). \tag{5.3}$$

The definition gives $W_a(0) = 0$, while the centering in (5.2) gives $W_a(1) = 0$.

For $0 < c < 1$, put $d := 1 - c$ and define

$$\mathcal{T}_c f := \mathrm{PV} \int_0^1 \frac{f(x)}{x-c}\,\mathrm{d}x + \pi\sqrt{\frac{c}{d}}\,f(c), \tag{5.4}$$

whenever the principal value exists. The main task of this section is to prove

$$\mathcal{T}_c R_a > 0, \qquad 0 < c < 1.$$

5.1. **Boundary values of the kernel.** Define

$$G_a(t) := \sum_{k\ge 1} \frac{\dot\alpha_k(a)}{k} t^k, \qquad \gamma(t) := \frac{(1+t)^2}{4t}, \qquad 0 < |t| < 1, \tag{5.5}$$

where $\alpha_k(a)$ is given by (3.4). The Joukowski map converts the kernel into a Stieltjes transform with controlled boundary values.

**Proposition 5.1.** *Let $a \ge 1$. Let $H_a$ be defined by (3.17), and let $W_a$, $R_a$, $G_a$, and $\gamma$ be defined by (5.3) and (5.5). Then*

$$H_a(t) = \frac{t-1}{2t} \int_0^1 \frac{R_a(x)}{x - \gamma(t)}\,\mathrm{d}x, \qquad 0 < |t| < 1. \tag{5.6}$$

*If $0 < \theta < \pi$ and $c := \cos^2\frac{\theta}{2}, d := \sin^2\frac{\theta}{2}$, then*

$$\operatorname{Re} H_a(\mathrm{e}^{i\theta}) = d\,\mathcal{T}_c R_a, \tag{5.7}$$

*where $\mathcal{T}_c$ is defined by (5.4).*

*Proof.* Using (3.3) and (3.4), followed by the substitution $x = \cos^2(\phi/2)$, we obtain

$$\alpha_k(a) = \int_0^1 \omega_a(x)\,\mathrm{T}_k(2x-1)\,\mathrm{d}x,$$

where $\mathrm{T}_k$ is the $k$th Chebyshev polynomial. Since $W_a' = -\omega_a(\log x - m_a)$ and $W_a$ vanishes at both endpoints, differentiation under the integral sign and then integration by parts yield

$$\dot\alpha_k(a) = 2k \int_0^1 W_a(x)\,\mathrm{U}_{k-1}(2x-1)\,\mathrm{d}x. \tag{5.8}$$

Here $\mathrm{U}_k$ denotes the Chebyshev polynomial of the second kind. We use the generating function

$$\sum_{j\ge 0} \mathrm{U}_j(x) t^j = \frac{1}{1 - 2xt + t^2}, \qquad |t| < 1,$$

see [OLBC10, p. 449, Eq. (18.12.10)].

The integrable majorant $x^{a-1/2}(1-x)^{-1/2}(1 + |\log x|)$ justifies differentiation under the integral sign. Moreover,

$$|\mathrm{U}_j(s)| \le j+1, \qquad -1 \le s \le 1,$$

and $W_a \in L^1(0,1)$. Hence, for every fixed $|t| < 1$,

$$\sum_{j\ge 0} |t|^{j+1} \int_0^1 |W_a(x)|\,|\mathrm{U}_j(2x-1)|\,\mathrm{d}x \le \|W_a\|_{L^1(0,1)} \sum_{j\ge 0} (j+1)|t|^{j+1} < \infty.$$

Thus Tonelli's theorem [Fol99, p. 67, Theorem 2.37(a)], applied to the absolute values, shows that the resulting series is absolutely integrable; Fubini's theorem [Fol99, p. 67, Theorem 2.37(b)] then permits summation of (5.8) under the integral sign. Using the generating function of $\mathrm{U}_j$, we obtain

$$G_a(t) = \frac{1}{2} \int_0^1 \frac{W_a(x)}{\gamma(t) - x}\,\mathrm{d}x. \tag{5.9}$$

From (5.5) and (3.17),

$$(1+t)G_a'(t) + a(t^{-1}-1)G_a(t) = \sum_{k\geq 1} \frac{\dot{\alpha}_k(a)}{k} \Big((a+k)t^{k-1} - (a-k)t^k\Big)$$
$$= H_a(t).$$

Set

$$I_1(t) := \int_0^1 \frac{W_a(x)}{\gamma(t)-x}\,\mathrm{d}x, \qquad I_2(t) := \int_0^1 \frac{W_a(x)}{(\gamma(t)-x)^2}\,\mathrm{d}x.$$

By (5.9),

$$G_a(t) = \frac{1}{2} I_1(t), \qquad G_a'(t) = -\frac{\gamma'(t)}{2} I_2(t).$$

Since $-(1+t)\gamma'(t) = \frac{1-t}{t}\gamma(t)$, we obtain

$$H_a(t) = \frac{1-t}{2t}\left(aI_1(t) + \gamma(t)I_2(t)\right).$$

On the other hand, integration by parts, together with $W_a(0) = W_a(1) = 0$, gives

$$\int_0^1 \frac{xW_a'(x)}{\gamma(t)-x}\,\mathrm{d}x = -\gamma(t)I_2(t).$$

Therefore

$$\int_0^1 \frac{R_a(x)}{x-\gamma(t)}\,\mathrm{d}x = -aI_1(t) - \gamma(t)I_2(t),$$

and hence

$$H_a(t) = \frac{t-1}{2t}\int_0^1 \frac{R_a(x)}{x-\gamma(t)}\,\mathrm{d}x.$$

This proves (5.6).

Now write $t = r\mathrm{e}^{i\theta}$, where $0 < r < 1$ and $0 < \theta < \pi$. Since $\gamma(t) = \frac{1}{2} + \frac{1}{4}\left(t + t^{-1}\right)$, we have

$$\Im\gamma(t) = \frac{r - r^{-1}}{4}\sin\theta < 0.$$

Thus, as $r \uparrow 1$, the point $\gamma(t)$ tends to $c = \cos^2\frac{\theta}{2}$ from the lower half-plane. The function $R_a$ is $C^1$ on compact subintervals of $(0,1)$. Moreover, the definitions of $W_a$ and $R_a$ give

$$R_a(x) = O\Big(x^{a+1/2}(1+|\log x|)\Big) \quad (x\downarrow 0), \qquad R_a(x) = O\Big((1-x)^{-1/2}\Big) \quad (x\uparrow 1).$$

Hence the endpoint contributions are absolutely integrable. Since $x - \gamma(r\mathrm{e}^{i\theta})$ tends to $(x-c)+i0$, the Sokhotski–Plemelj formula [Mus58, pp. 42–44], applied after inserting a cutoff supported near $c$, and ordinary dominated convergence on the complementary part, give

$$\int_0^1 \frac{R_a(x)}{x-\gamma(t)}\,\mathrm{d}x \longrightarrow \mathrm{PV}\int_0^1 \frac{R_a(x)}{x-c}\,\mathrm{d}x - i\pi R_a(c).$$

Since $\frac{\mathrm{e}^{i\theta}-1}{2\mathrm{e}^{i\theta}} = d + i\sqrt{cd}$, taking real parts in (5.6) gives (5.7). □

5.2. **The beta transform.** Define

$$L_a(c) := (1-c)\mathcal{T}_c\omega_a, \qquad \kappa_a := \frac{\pi}{Z_a} = \frac{\sqrt{\pi}\,\Gamma(a+1)}{\Gamma(a+1/2)}. \tag{5.10}$$

The transform of the basic beta density is explicitly positive.

**Lemma 5.2.** *Let $a \geq 1$, $0 < c < 1$, and $d = 1-c$. Let $\omega_a$ be the beta density defined by (5.1), let $\mathcal{T}_c$ be defined by (5.4), and let $L_a(c)$ and $\kappa_a$ be defined by (5.10). Then*

$$L_a(c) = \kappa_a c^a + 2ad\int_0^1 (c+dt^2)^{a-1}\,\mathrm{d}t > 0. \tag{5.11}$$

*Moreover,*

$$\kappa_a \leq 2a, \tag{5.12}$$

*with equality only at $a = 1$.*

*Proof.* Let $U$ have density $au^{a-1}\mathbf{1}_{(0,1)}(u)$. Conditional on $U = u$, let $X$ have the shifted arcsine density

$$f_u(x) := \frac{\mathbf{1}_{(u,1)}(x)}{\pi\sqrt{(x-u)(1-x)}}.$$

For $0 < x < 1$, the beta integral gives

$$\int_0^1 au^{a-1}f_u(x)\,\mathrm{d}u = \frac{x^{a-1/2}(1-x)^{-1/2}}{Z_a} = \omega_a(x).$$

Thus the marginal density of $X$ is $\omega_a$.

We first take $z \in \mathbb{C}\setminus[0,1]$ as the Cauchy-transform parameter. Fubini's theorem then applies to the absolutely convergent double integral. The substitution $x = u + (1-u)\sin^2 v$ gives the Cauchy transform

$$\int_u^1 \frac{f_u(x)}{x-z}\,\mathrm{d}x = \frac{1}{\sqrt{(u-z)(1-z)}},$$

with the branch positive for real $z < u$. In the boundary passage below we approach $c$ from the lower half-plane. Thus, when $u < c$,

$$\lim_{\varepsilon\downarrow 0}\int_u^1 \frac{f_u(x)}{x-(c-i\varepsilon)}\,\mathrm{d}x = \mathrm{PV}\int_u^1 \frac{f_u(x)}{x-c}\,\mathrm{d}x - i\pi f_u(c).$$

This fixes the Sokhotski–Plemelj sign convention used below. We now distinguish the cases $u > c$ and $u < c$.

If $u > c$, then $c$ lies outside the support of $f_u$, and hence

$$\mathcal{T}_c f_u = \int_u^1 \frac{f_u(x)}{x-c}\,\mathrm{d}x = \frac{1}{\sqrt{(u-c)d}}.$$

If $u < c$, the two boundary values of $\frac{1}{\sqrt{(u-z)(1-z)}}$ at $z = c$ are purely imaginary. Therefore

$$\mathrm{PV}\int_u^1 \frac{f_u(x)}{x-c}\,\mathrm{d}x = 0.$$

Moreover,

$$f_u(c) = \frac{1}{\pi\sqrt{(c-u)d}},$$

so the second term in (5.4) gives

$$\pi\sqrt{\frac{c}{d}}\,f_u(c) = \frac{\sqrt{c}}{d\sqrt{c-u}}.$$

Consequently, for $u \neq c$,

$$d\,\mathcal{T}_c f_u = \sqrt{\frac{c}{c-u}}\,\mathbf{1}_{\{u<c\}} + \sqrt{\frac{d}{u-c}}\,\mathbf{1}_{\{u>c\}}.$$

For $z = c - i\varepsilon$ in a fixed sufficiently small neighborhood of $c$, the preceding Cauchy kernels satisfy

$$\left|\frac{1}{\sqrt{(u-z)(1-z)}}\right| \le C_c|u-c|^{-1/2}.$$

Since $au^{a-1}|u-c|^{-1/2} \in L^1(0,1)$, dominated convergence on the mixture side gives the lower boundary value of the Cauchy transform of $\omega_a$. On the other hand, $\omega_a$ is $C^1$ in a neighborhood of $c$ and is integrable at the endpoints. Hence a cutoff argument and the Sokhotski–Plemelj formula [Mus58, pp. 42–44] give

$$\lim_{\varepsilon\downarrow 0}\int_0^1 \frac{\omega_a(x)}{x-(c-i\varepsilon)}\,\mathrm{d}x = \mathrm{PV}\int_0^1 \frac{\omega_a(x)}{x-c}\,\mathrm{d}x - i\pi\omega_a(c).$$

Comparing real parts with the boundary value of the mixture representation yields

$$\mathrm{PV}\int_0^1 \frac{\omega_a(x)}{x-c}\,\mathrm{d}x = a\int_c^1 \frac{u^{a-1}}{\sqrt{(u-c)d}}\,\mathrm{d}u.$$

On the other hand, Tonelli's theorem applied to the marginal identity at $x = c$ gives

$$\omega_a(c) = a\int_0^c u^{a-1} f_u(c)\,\mathrm{d}u.$$

Adding $\pi\sqrt{c/d}\,\omega_a(c)$ to the preceding principal-value identity therefore yields

$$d\,\mathcal{T}_c\omega_a = a\int_0^1 u^{a-1}\,d\,\mathcal{T}_c f_u\,\mathrm{d}u.$$

Consequently,

$$\begin{aligned} L_a(c) &= a\sqrt{c}\int_0^c \frac{u^{a-1}}{\sqrt{c-u}}\,\mathrm{d}u + a\sqrt{d}\int_c^1 \frac{u^{a-1}}{\sqrt{u-c}}\,\mathrm{d}u \\ &= \kappa_a c^a + 2ad\int_0^1 (c+dt^2)^{a-1}\,\mathrm{d}t. \end{aligned}$$

The first integral is a beta integral, and the second equality uses $u = c + dt^2$. This proves (5.11).

For (5.12), note that $\kappa_1 = 2$ and

$$\frac{\mathrm{d}}{\mathrm{d}a}\log\frac{\kappa_a}{a} = \psi(a+1) - \psi\left(a+\frac{1}{2}\right) - \frac{1}{a}.$$

The digamma integral formula [OLBC10, p. 140, Eq. (5.9.16)] gives

$$\psi(a+1) - \psi\left(a+\frac{1}{2}\right) = \int_0^\infty \frac{\mathrm{e}^{-at}}{\mathrm{e}^{t/2}+1}\,\mathrm{d}t < \frac{1}{2a}.$$

Thus $a \mapsto \kappa_a/a$ is strictly decreasing, which proves the claim. $\square$

5.3. **A positive Abel decomposition.** The function $R_a$ need not have a fixed sign. The following positive Abel decomposition resolves it into elementary pieces whose transforms can be controlled individually.

**Lemma 5.3.** *Let $a \geq 1$. Let $Z_a$ be defined by (5.1), and let $W_a$ and $R_a$ be defined by (5.3). There are numbers*

$$\delta_N(a) > 0, \qquad \sum_{N \geq 0} \delta_N(a) < \infty, \tag{5.13}$$

*such that*

$$W_a(x) = \frac{1}{Z_a} \sum_{N \geq 0} \delta_N(a) F_{a,N}(x), \tag{5.14}$$

*where*

$$F_{a,N}(x) := x^{a-1/2}(1-x)^{1/2}\big(1-(1-x)^{N+1}\big). \tag{5.15}$$

*If*

$$R_{a,N}(x) := aF_{a,N}(x) - xF'_{a,N}(x), \tag{5.16}$$

*then*

$$R_a(x) = \frac{1}{Z_a} \sum_{N \geq 0} \delta_N(a) R_{a,N}(x). \tag{5.17}$$

*The series in (5.17) converges absolutely in $L^1(0,1)$. The series in (5.14) and (5.17), together with their termwise derivatives, converge locally uniformly on $(0,1)$.*

*Proof.* Write $y = 1-x$. Since the centered integrand in (5.3) has integral zero,

$$W_a(1-y) = \frac{1}{Z_a} \int_0^y (1-v)^{a-1/2} v^{-1/2} \big(\log(1-v) - m_a\big)\, \mathrm{d}v.$$

Set $\Phi_a(v) := (1-v)^{a-1/2}\big(\log(1-v) - m_a\big)$. The branch determined at $v = 0$ is analytic for $|v| < 1$. If $\Phi_a(v) = \sum_{j \geq 0} b_j v^j$, then termwise integration gives, for $0 < y < 1$,

$$\int_0^y v^{-1/2} \Phi_a(v)\, \mathrm{d}v = y^{1/2} \sum_{j \geq 0} \frac{b_j}{j+1/2} y^j.$$

The series on the right is analytic for $|y| < 1$. Multiplication by the branch of $(1-y)^{-a-1/2}$ that equals 1 at $y = 0$ therefore gives an analytic function $\tau_a$ on the unit disk such that

$$W_a(x) = \frac{1}{Z_a} x^{a+1/2} y^{1/2} \tau_a(y), \qquad \tau_a(y) = \sum_{j \geq 0} t_j y^j. \tag{5.18}$$

Substitution into $W'_a = -\omega_a(\log x - m_a)$ gives

$$y(1-y)\tau'_a(y) + \left(\frac{1}{2} - (a+1)y\right) \tau_a(y) = \log(1-y) - m_a.$$

Comparing coefficients yields

$$t_0 = -2m_a, \qquad t_j = \frac{a+j}{j+1/2} t_{j-1} - \frac{1}{j(j+1/2)} \quad (j \geq 1). \tag{5.19}$$

Motivated by the backward recurrence, define

$$\widetilde{t}_j := \sum_{k=j+1}^{\infty} \left( \prod_{\ell=j+1}^{k-1} \frac{\ell+1/2}{a+\ell} \right) \frac{1}{k(a+k)}. \tag{5.20}$$

Every term is positive, and comparison with $\sum_{k>j} k^{-2}$ gives absolute convergence and $\widetilde{t}_j \to 0$. The sequence $(\widetilde{t}_j)$ satisfies the backward form of (5.19). To identify its initial value, let $Y \sim \mathrm{Beta}(1/2, a+1/2)$. Since $1-Y$ has density $\omega_a$ defined in (5.1),

$$\widetilde{t}_0 = 2\sum_{k\geq 1} \frac{(1/2)_k}{(a+1)_k}\frac{1}{k} = 2\mathbb{E}[-\log(1-Y)] = -2m_a = t_0.$$

Here the exchange of expectation and series follows from monotone convergence applied to

$$-\log(1-Y) = \sum_{k\geq 1} \frac{Y^k}{k}.$$

Since the forward recurrence (5.19) uniquely determines its solution from $t_0$, we have $\widetilde{t}_j = t_j$ for every $j \geq 0$. Thus (5.20) gives the coefficients in (5.18), and in particular $t_j \to 0$.

The backward form of (5.19) is

$$t_j = \frac{j+3/2}{a+j+1} t_{j+1} + \frac{1}{(j+1)(a+j+1)}. \tag{5.21}$$

For $j \geq 1$, set $E_j := 1/[j(a-1/2)]$. For $K \geq 1$, define

$$t_j^{(K)} := \sum_{k=j+1}^{K} \left( \prod_{\ell=j+1}^{k-1} \frac{\ell + 1/2}{a+\ell} \right) \frac{1}{k(a+k)}, \qquad 0 \leq j \leq K,$$

so that $t_K^{(K)} = 0$. Iterating (5.21) downward, one obtains

$$t_{j+1}^{(K)} \leq E_{j+1} \quad \Longrightarrow \quad t_j^{(K)} \leq \frac{j+3/2}{a+j+1} E_{j+1} + \frac{1}{(j+1)(a+j+1)} = E_{j+1}.$$

Applying the same formula once at $j=0$ gives $t_0^{(K)} \leq E_1$. Letting $K \to \infty$ therefore yields

$$0 < t_j \leq E_{j+1}, \qquad j \geq 0, \qquad t_j \longrightarrow 0. \tag{5.22}$$

Define

$$\delta_j(a) := t_j - t_{j+1}. \tag{5.23}$$

Equations (5.21) and (5.22) give

$$\delta_j(a) = \frac{1}{(j+1)(a+j+1)} - \frac{a-1/2}{a+j+1} t_{j+1} > 0,$$

because $t_{j+1} \leq E_{j+2} < [(j+1)(a-1/2)]^{-1}$. Since $t_j \to 0$, Abel summation gives

$$\tau_a(y) = \sum_{N\geq 0} \delta_N(a) \sum_{j=0}^{N} y^j, \qquad \sum_{N\geq 0} \delta_N(a) = t_0. \tag{5.24}$$

Substituting (5.24) into (5.18) proves (5.14).

A direct differentiation of (5.15) gives

$$R_{a,N}(x) = x^{a-1/2}(1-x)^{-1/2} r_N(1-x), \tag{5.25}$$

where

$$r_N(y) := \frac{1}{2} - \left(N + \frac{3}{2}\right) y^{N+1} + (N+1)y^{N+2}. \tag{5.26}$$

Since $r_N(y) = \frac{1}{2}(1-y^{N+1}) - (N+1)(1-y)y^{N+1}$, we have $|r_N(y)| \leq 3/2$ on $[0,1]$. Together with (5.13), this proves absolute $L^1$ convergence in (5.17). Let $I \Subset (0,1)$. The quantities $N^2(1-x)^N$ are bounded uniformly for $N \geq 0$ and $x \in I$. The explicit formulas (5.15) and (5.25) therefore give uniform bounds on $I$ for $F_{a,N}$, $F'_{a,N}$, $R_{a,N}$, and $R'_{a,N}$. Since $\sum_N \delta_N(a) < \infty$, the Weierstrass $M$-test proves the asserted local uniform convergence. □

5.4. **The finite-basis inequality.** For every integer $m \geq 0$, define the beta moments

$$\nu_m(a) := \int_0^1 \omega_a(x)(1-x)^m \, \mathrm{d}x = \frac{(1/2)_m}{(a+1)_m}.$$

They satisfy

$$(a+m)\nu_m(a) = \left(m - \frac{1}{2}\right)\nu_{m-1}(a), \qquad m \geq 1. \tag{5.27}$$

We next prove that the transform is positive on every basis element in the Abel decomposition.

**Lemma 5.4.** *Let $a \geq 1$, let $N \geq 0$ be an integer, and let $0 < c < 1$. Let $Z_a$ be defined by (5.1), let $\mathcal{T}_c$ be the transform in (5.4), and let $R_{a,N}$ be defined by (5.16). Set*

$$\mathcal{J}_{a,N}(c) := \frac{1}{Z_a}\mathcal{T}_c R_{a,N}. \tag{5.28}$$

*Then*

$$\mathcal{J}_{a,N}(c) > 0. \tag{5.29}$$

*Proof.* Put $d = 1 - c$. For every $m \geq 0$, polynomial division gives

$$\frac{(1-x)^m}{x-c} = \frac{d^m}{x-c} - \sum_{k=0}^{m-1} d^k (1-x)^{m-1-k},$$

where the sum is empty for $m = 0$. Applying $\mathcal{T}_c$ and using (5.10) yields

$$\mathcal{T}_c\big[\omega_a(x)(1-x)^m\big] = d^{m-1}L_a(c) - \sum_{k=0}^{m-1} d^k \nu_{m-1-k}(a), \tag{5.30}$$

where the sum is empty when $m = 0$ and $L_a$ is defined by (5.10).

By (5.25)–(5.26),

$$Z_a^{-1}R_{a,N}(x) = \omega_a(x) r_N(1-x).$$

Hence

$$\begin{aligned}\mathcal{J}_{a,N}(c) &= \frac{1}{2}\,\mathcal{T}_c\omega_a - \left(N + \frac{3}{2}\right)\mathcal{T}_c\big[\omega_a(x)(1-x)^{N+1}\big] \\ &\quad + (N+1)\mathcal{T}_c\big[\omega_a(x)(1-x)^{N+2}\big].\end{aligned}$$

Applying (5.30), the coefficient of $L_a(c)$ is

$$\frac{1}{2d} - \left(N + \frac{3}{2}\right)d^N + (N+1)d^{N+1} = \frac{1}{2d} - \left(\frac{1}{2} + c(N+1)\right)d^N.$$

For the remaining terms, (5.27) gives

$$\begin{aligned}&\left(N + \frac{3}{2}\right)\sum_{k=0}^{N} d^k \nu_{N-k}(a) - (N+1)\sum_{k=0}^{N+1} d^k \nu_{N+1-k}(a) \\ &\quad = a\sum_{k=0}^{N} d^k \nu_{N+1-k}(a) + c\sum_{k=1}^{N+1} k d^{k-1}\nu_{N+1-k}(a).\end{aligned}$$

Consequently,

$$\mathcal{J}_{a,N}(c) = L_a(c)C_N(c) + aU_N(a,c) + cV_N(a,c),$$

where

$$C_N(c) := \frac{1}{2d} - \left(\frac{1}{2} + c(N+1)\right) d^N, \tag{5.31}$$

$$U_N(a,c) := \sum_{k=0}^{N} d^k \nu_{N+1-k}(a), \qquad V_N(a,c) := \sum_{k=1}^{N+1} k d^{k-1} \nu_{N+1-k}(a). \tag{5.32}$$

In particular, $U_N$ and $V_N$ are positive.

*The base case.* From (5.26),

$$R_{a,0}(x) = x^{a-1/2}(1-x)^{-1/2} x \left(x - \frac{1}{2}\right).$$

Since $x\omega_a(x) = \frac{a+1/2}{a+1}\,\omega_{a+1}(x)$,

$$\mathcal{J}_{a,0}(c) = \frac{a+1/2}{a+1}\left[1 + \left(c - \frac{1}{2}\right)\frac{L_{a+1}(c)}{1-c}\right]. \tag{5.33}$$

This is positive when $c \geq 1/2$ because $L_{a+1}(c) > 0$ by Lemma 5.2.

Suppose $0 < c < 1/2$ and set $A = a+1 > 1$. Since $t^2 \leq t$ on $[0,1]$, Lemma 5.2 and (5.12) give

$$L_A(c) \leq \kappa_A c^A + 2Ad \int_0^1 (c+dt)^{A-1}\, \mathrm{d}t \leq 2 + 2(A-1)c^A. \tag{5.34}$$

The maximum of $(A-1)c^{A-1}(1-2c)$ on $(0,1/2)$ is $2^{1-A}(1-1/A)^A < 1$. Hence

$$(A-1)c^A < \frac{c}{1-2c}, \qquad 2 + 2(A-1)c^A < \frac{2(1-c)}{1-2c}.$$

Thus $L_A(c) < 2d/(1-2c)$, and (5.33) again gives

$$\mathcal{J}_{a,0}(c) > 0. \tag{5.35}$$

*The induction step.* Termwise subtraction in (5.31)–(5.32) gives

$$\begin{aligned} U_{N+1} - dU_N &= \nu_{N+2}(a), \\ V_{N+1} - dV_N &= U_N + d^{N+1}, \\ C_{N+1} - dC_N &= c\left(\frac{1}{2d} - d^{N+1}\right). \end{aligned}$$

With $m = N+1$ and $\Sigma_m(a,c) := \sum_{k=0}^m d^k \nu_{m-k}(a)$, we obtain

$$\mathcal{J}_{a,N+1}(c) - d\mathcal{J}_{a,N}(c) = a\nu_{m+1}(a) + c\Sigma_m(a,c) + cL_a(c)\left(\frac{1}{2d} - d^m\right). \tag{5.36}$$

If $d^{m+1} \leq 1/2$, every term on the right is nonnegative and the first is positive.

Assume now that

$$\xi := d^{m+1} > \frac{1}{2}, \qquad \rho := 1 - \frac{1}{2\xi} \in \left(0, \frac{1}{2}\right).$$

Then $d > 2^{-1/(m+1)} > 1/2$, so $c < 1/2$. The last term in (5.36) is negative. After moving it to the other side and dividing by $cd^m$, it is enough to prove

$$L_a(c)\rho < \frac{\Sigma_m(a,c)}{d^m} + \frac{a\nu_{m+1}(a)}{cd^m}. \tag{5.37}$$

The right-hand side is strictly larger than 1. Indeed, the term $k = m$ in $\Sigma_m(a,c)/d^m$ equals $\frac{d^m \nu_0(a)}{d^m} = 1$, and all the remaining terms, including $a\nu_{m+1}(a)/(cd^m)$, are strictly positive. On the other hand, the same estimate as in (5.34) gives

$$L_a(c) \leq 2 + 2(a-1)c^a. \tag{5.38}$$

Furthermore,

$$\frac{1-\xi}{\xi} = \frac{c(1+d+\cdots+d^m)}{d^{m+1}} \geq \frac{c}{d}.$$

Since $c < 1/2$, one has $e|\log c| > 1$; moreover, $d < 1$. If $a > 1$, then, with $s = (a-1)|\log c|$,

$$(a-1)c^{a-1} = \frac{se^{-s}}{|\log c|} \leq \frac{1}{e|\log c|} < 1 < \frac{1}{d},$$

so

$$(a-1)c^a < \frac{c}{d} \leq \frac{1-\xi}{\xi}. \tag{5.39}$$

Set $y := (1-\xi)/\xi \in (0,1)$. Then $2\rho = 1-y$. For $a > 1$, equations (5.38) and (5.39) give

$$L_a(c)\rho < (2+2y)\frac{1-y}{2} = 1-y^2 < 1.$$

When $a = 1$, formula (5.11) gives $L_1(c) = 2$, and hence $L_1(c)\rho = 1-y < 1$. Thus (5.37) holds in all cases, and

$$\mathcal{J}_{a,N+1}(c) > d\mathcal{J}_{a,N}(c).$$

Together with (5.35), this proves (5.29) by induction. □

5.5. **Passage to the principal value.** The preceding estimates allow us to pass to the principal value and complete the range $a \geq 1$.

**Proposition 5.5.** *Let $a \geq 1$. Let $\mathcal{T}_c$, $R_a$, $\delta_N(a)$, $\mathcal{J}_{a,N}(c)$, and $H_a$ be defined by (5.4), (5.3), (5.23), (5.28), and (3.17), respectively. Then, for every $0 < c < 1$,*

$$\mathcal{T}_c R_a = \sum_{N\geq 0} \delta_N(a)\mathcal{J}_{a,N}(c) > 0. \tag{5.40}$$

*Consequently,*

$$\operatorname{Re} H_a(z) > 0, \qquad z \in \mathbb{D}. \tag{5.41}$$

*Proof.* For a function $f \in L^1(0,1)$ that is locally Lipschitz near $c$,

$$\mathrm{PV}\int_0^1 \frac{f(x)}{x-c}\,\mathrm{d}x = \int_0^1 \frac{f(x)-f(c)}{x-c}\,\mathrm{d}x + f(c)\log\frac{1-c}{c}. \tag{5.42}$$

Choose a closed interval $I \Subset (0,1)$ whose interior contains $c$. From (5.25)–(5.26), there is a constant $C_I$ such that

$$\|R_{a,N}\|_{L^\infty(I)} + \left\|R'_{a,N}\right\|_{L^\infty(I)} \leq C_I \qquad (N \geq 0).$$

For $x \in I$, the mean-value theorem gives

$$\left|\frac{R_{a,N}(x) - R_{a,N}(c)}{x-c}\right| \leq C_I, \qquad N \geq 0.$$

Outside $I$, the denominator is bounded away from zero. Moreover, (5.25) and $|r_N| \leq 3/2$ give

$$|R_{a,N}(x)| \leq \frac{3}{2}x^{a-1/2}(1-x)^{-1/2}, \qquad 0 < x < 1,$$

uniformly in $N$. The preceding local uniform bound on $I$ also gives

$$\sup_{N\geq 0} |R_{a,N}(c)| < \infty.$$

It follows that

$$\left|\frac{R_{a,N}(x) - R_{a,N}(c)}{x-c}\right|$$

has an integrable majorant on $(0,1)\setminus I$ that is independent of $N$. Together with the mean-value bound on $I$, this justifies termwise passage in the difference-quotient term in (5.42). The preceding bounds imply

$$\sup_{N\geq 0}\int_0^1 \left|\frac{R_{a,N}(x)-R_{a,N}(c)}{x-c}\right| \,\mathrm{d}x<\infty, \qquad \sup_{N\geq 0}|R_{a,N}(c)|<\infty.$$

Since $\sum_{N\geq 0}\delta_N(a)<\infty$, Fubini's theorem applies to the difference-quotient series, while local uniform convergence handles the value and residue terms. Hence

$$\mathcal{T}_c R_a=\frac{1}{Z_a}\sum_{N\geq 0}\delta_N(a)\mathcal{T}_c R_{a,N}=\sum_{N\geq 0}\delta_N(a)\mathcal{J}_{a,N}(c).$$

The right-hand side is strictly positive by Lemmas 5.3 and 5.4. This proves (5.40).

Proposition 5.1 now gives $\operatorname{Re} H_a(\mathrm{e}^{i\theta})>0$ for $0<\theta<\pi$. Since the Taylor coefficients of $H_a$ are real, the same holds on the lower open semicircle.

It remains to treat the two endpoints. First let $t=-r$ and $r\uparrow 1$ in (5.6). Then $\gamma(-r)=-(1-r)^2/(4r)\uparrow 0$. From (5.3),

$$\begin{aligned}
W_a(x)&=O\big(x^{a+1/2}(1+|\log x|)\big), & W_a'(x)&=O\big(x^{a-1/2}(1+|\log x|)\big), \quad x\downarrow 0,\\
W_a(x)&=O((1-x)^{1/2}), & W_a'(x)&=O((1-x)^{-1/2}), \quad x\uparrow 1.
\end{aligned}$$

Hence $R_a(x)/x\in L^1(0,1)$, and dominated convergence applies in (5.6). Since $W_a(0)=W_a(1)=0$, $\int_0^1 W_a'(x)\,\mathrm{d}x=0$, and therefore

$$H_a(-1)=\int_0^1\frac{R_a(x)}{x}\,\mathrm{d}x=a\int_0^1\frac{W_a(x)}{x}\,\mathrm{d}x=a\int_0^1\omega_a(x)(\log x-m_a)^2\,\mathrm{d}x>0.$$

The last identity follows from Fubini's theorem: absolute integrability is ensured by $\omega_a(x)(1+|\log x|^2)\in L^1(0,1)$, and (5.2) gives

$$\int_0^1\frac{W_a(x)}{x}\,\mathrm{d}x=\int_0^1\omega_a(u)(\log u-m_a)\log u\,\mathrm{d}u.$$

At the other endpoint, direct evaluation of (3.3) and the duplication formula give $h_a(1)=\kappa_a$. The differentiated Fourier series used in Proposition 3.2 converges absolutely and uniformly. Evaluating it at 1 gives

$$\dot h_a(1)=2\sum_{k\geq 1}\dot\alpha_k(a).$$

On the other hand, (3.17) gives

$$\begin{aligned}
H_a(1)&=\sum_{k\geq 1}\frac{\dot\alpha_k(a)}{k}\big((a+k)-(a-k)\big)\\
&=2\sum_{k\geq 1}\dot\alpha_k(a)=\dot h_a(1).
\end{aligned}$$

Therefore

$$\begin{aligned}
H_a(1)&=\frac{\mathrm{d}}{\mathrm{d}a}h_a(1)\\
&=\kappa_a\left[\psi(a+1)-\psi\left(a+\frac12\right)\right]>0.
\end{aligned}$$

Thus $\operatorname{Re} H_a$ is harmonic in $\mathbb{D}$, continuous on $\overline{\mathbb{D}}$, and positive on $\mathbb{T}$. The minimum principle gives (5.41) throughout $\mathbb{D}$. □

Combining Proposition 5.5 with Section 4 completes the proof of Theorem 3.5.

## 6. Completion of the proof

We first assemble the preceding reductions under the simple-zero hypothesis, and then remove that hypothesis by approximation.

**Proposition 6.1.** *Let $p$ satisfy the normalization (2.3), assume that its zeros on $\mathbb{T}$ are simple, and set $Q_n(z) := 1 + z^n$. Then*

$$F_p(r) := \log \frac{\|p\|_r}{\|Q_n\|_r} \tag{6.1}$$

*is nondecreasing for $r > 0$.*

*Proof.* Fix $r = 2a > 0$. Theorem 3.5 and the area identity (3.18) give $\mathcal{P}_{2a}(q, B) \geq 0$. Proposition 2.4 therefore yields

$$\mathsf{D}(\rho_{p,r} \| m) \geq \mathsf{D}(\rho_{Q_n,r} \| m).$$

Lemma 2.3 now gives $F_p'(r) \geq 0$. □

To remove the simple-zero hypothesis, we perturb the arguments of the roots while preserving their product.

**Lemma 6.2.** *Let $p$ satisfy (2.3), with arbitrary multiplicities among its zeros on $\mathbb{T}$. Then there are polynomials $p_\nu$ satisfying the same normalization, each with simple zeros on $\mathbb{T}$, such that $p_\nu \to p$ coefficientwise and uniformly on $\mathbb{T}$.*

*Proof.* List the zeros with multiplicity as $\mathrm{e}^{i\theta_1}, \ldots, \mathrm{e}^{i\theta_n}$, choosing the same real representative for repeated zeros. For $n \geq 2$, set

$$v_j := j - \frac{n+1}{2}, \qquad \sum_{j=1}^{n} v_j = 0.$$

Let $\delta > 0$ be the least circular distance between two distinct original zeros, with $\delta = 2\pi$ if there is only one distinct zero. Choose $\varepsilon_\nu \downarrow 0$ so that $\varepsilon_\nu(n-1) < \min\{\delta/2, 2\pi\}$, and define

$$p_\nu(z) := \prod_{j=1}^{n} \left( z - \mathrm{e}^{i(\theta_j + \varepsilon_\nu v_j)} \right).$$

If $\theta_j = \theta_k$ and $j \neq k$, then $0 < \varepsilon_\nu |v_j - v_k| < 2\pi$, so the corresponding perturbed roots are distinct. If the two original zeros represented by $\theta_j$ and $\theta_k$ are distinct, then each corresponding perturbed root moves by at most $\varepsilon_\nu(n-1)/2 < \delta/4$ in circular distance. Hence roots arising from two distinct original clusters remain separated by more than $\delta/2$. Thus all zeros of $p_\nu$ are simple and lie on $\mathbb{T}$. Moreover, $\sum_j v_j = 0$, so the product of the roots, and therefore the constant coefficient, is unchanged. The polynomials $p_\nu$ are consequently monic with constant coefficient 1. Since all their zeros are unimodular, they also satisfy $p_\nu = p_\nu^*$; thus they have the normalization (2.3). Their roots converge to those of $p$ with multiplicity, so $p_\nu \to p$ coefficientwise. Because the degree is fixed, the convergence is uniform on $\mathbb{T}$. For $n = 1$, no perturbation is required. □

We can now prove Theorem 1.1.

*Proof.* First suppose that $p$ satisfies (2.3). Let $p_\nu$ be the approximants from Lemma 6.2. For every fixed $0 < r < \infty$, uniform convergence on $\mathbb{T}$ gives $\|p_\nu\|_r \to \|p\|_r$. Fix $0 < s < t < \infty$. Proposition 6.1 gives

$$\log \frac{\|p_\nu\|_s}{\|Q_n\|_s} \leq \log \frac{\|p_\nu\|_t}{\|Q_n\|_t}.$$

Passing to the limit $\nu \to \infty$ yields the same inequality for $p$. Hence $F_p$ in (6.1) is nondecreasing on $(0, \infty)$.

It remains to include the endpoints. Jensen's formula gives

$$\|p\|_0 = \|Q_n\|_0 = 1.$$

To see continuity at 0, define $X(\zeta) = \log|p(\zeta)|$ when $p(\zeta) \neq 0$, and set $X(\zeta) = 0$ on the finite zero set of $p$. Then $X$ is bounded above, belongs to $L^1(\mathbb{T}, m)$, and $\mathrm{e}^{rX} = |p|^r$ almost everywhere. For $0 < r \leq 1$,

$$\left|\frac{\mathrm{e}^{rX} - 1}{r}\right| \leq |X| \quad \text{on } \{X < 0\}, \qquad \left|\frac{\mathrm{e}^{rX} - 1}{r}\right| \leq \mathrm{e}^{\sup X}|X| \quad \text{on } \{X \geq 0\}.$$

The finite zero set is null, so the dominated convergence theorem gives

$$\int_{\mathbb{T}} \mathrm{e}^{rX}\, \mathrm{d}m = 1 + r\int_{\mathbb{T}} X\, \mathrm{d}m + o(r), \qquad r \downarrow 0.$$

Thus $M_p(r) = 1 + r\int_{\mathbb{T}} X\, \mathrm{d}m + o(r)$. After taking logarithms and dividing by $r$, we obtain $\log\|p\|_r \to \int_{\mathbb{T}} \log|p|\, \mathrm{d}m$, and hence $\|p\|_r \to \|p\|_0$. The same argument applies to $Q_n$. At the other endpoint, the usual $L^r$ limit on the probability space $(\mathbb{T}, m)$ gives

$$\|p\|_r \longrightarrow \|p\|_\infty, \qquad \|Q_n\|_r \longrightarrow \|Q_n\|_\infty \quad (r \to \infty).$$

Taking the appropriate one-sided limits in the positive-exponent inequality now extends it to every pair $0 \leq s \leq t \leq \infty$.

Finally, let $P$ be the original polynomial. By Lemma 2.1, $p(z) = cP(\eta z)$ for some $c \neq 0$ and $\eta \in \mathbb{T}$. Rotation preserves every $L^r$ mean, while multiplication by $c$ scales all means by the same factor $|c|$. Therefore the ratio inequality for $p$ is equivalent to the ratio inequality for $P$. □

## 7. Consequences and neighboring extremal problems

For $0 < r < \infty$, set

$$C_r := \|1 + z\|_r, \qquad C_0 := 1, \qquad C_\infty := 2. \tag{7.1}$$

Since $z \mapsto z^n$ preserves the normalized Haar measure $m$, one has $\|Q_n\|_r = C_r$. The beta integral gives

$$C_r^r = \frac{1}{2\pi}\int_0^{2\pi} |1 + \mathrm{e}^{i\theta}|^r\, \mathrm{d}\theta = \frac{\Gamma(r+1)}{\Gamma(1 + r/2)^2}.$$

In particular,

$$C_1 = \frac{4}{\pi}, \qquad C_2 = \sqrt{2}, \qquad C_4 = 6^{1/4}.$$

### 7.1. Reverse norm and coefficient inequalities.

Theorem 1.1 first yields the following sharp family of norm and coefficient inequalities.

**Corollary 7.1.** *Let*

$$p(z) = \sum_{k=0}^{n} a_k z^k = a_n \prod_{j=1}^{n}(z - \zeta_j), \qquad |\zeta_j| = 1, \tag{7.2}$$

*be a nonzero polynomial of degree $n \geq 1$. Let $C_r$ be defined by (7.1).*

(i) *For every $0 \leq s \leq t \leq \infty$,*

$$\|p\|_t \geq \frac{C_t}{C_s}\|p\|_s. \tag{7.3}$$

*The constant $C_t/C_s$ is sharp.*

(ii) *For every $0 \leq r \leq \infty$,*

$$|a_0| + |a_n| \leq \frac{2}{C_r}\|p\|_r. \tag{7.4}$$

*Thus the constants for $r = 1, 2, \infty$ are $\pi/2$, $\sqrt{2}$, and $1$, respectively.*

(iii) *One has*

$$\|p\|_\infty^2 \ge 2\,\|p\|_2^2 = 2\sum_{k=0}^{n} |a_k|^2. \tag{7.5}$$

*More generally, if $m$ is a positive integer and $p(z)^m = \sum_{\ell=0}^{mn} b_\ell z^\ell$, then*

$$\|p\|_\infty^{2m} \ge \frac{4^m}{\binom{2m}{m}} \sum_{\ell=0}^{mn} |b_\ell|^2. \tag{7.6}$$

(iv) *For every integer $m \ge 1$,*

$$\int_{\mathbb{T}} |p|^{2m}\, \mathrm{d}m \ge \frac{\binom{2m}{m}}{2^m} \left( \int_{\mathbb{T}} |p|^2\, \mathrm{d}m \right)^m. \tag{7.7}$$

*For $m = 2$, this is the convolution inequality*

$$\sum_{\ell=0}^{2n} \left| \sum_{\substack{u+v=\ell \\ 0\le u,v\le n}} a_u a_v \right|^2 \ge \frac{3}{2} \left( \sum_{k=0}^{n} |a_k|^2 \right)^2. \tag{7.8}$$

*The constants in* (i)–(iv) *are attained by $p(z) = c(z^n - \omega)$ with $c \neq 0$ and $|\omega| = 1$.*

*Proof.* Equation (7.3) is Theorem 1.1, because $\|Q_n\|_r = C_r$.

For (7.4), Jensen's formula and (7.2) give

$$\|p\|_0 = |a_n|, \qquad |a_0| = |a_n|.$$

Apply (7.3) with $(s,t) = (0,r)$. The case $r = \infty$ agrees with Visser's classical inequality [Vis45, p. 279, Theorem 3].

Taking $(s,t) = (2,\infty)$ proves (7.5); this is the O'Hara–Rodriguez inequality [OR74, p. 333, Corollary 1]. Taking $(s,t) = (2m,\infty)$ and using $C_{2m}^{2m} = \binom{2m}{m}$ gives (7.6), because

$$\|p\|_{2m}^{2m} = \|p^m\|_2^2 = \sum_{\ell=0}^{mn} |b_\ell|^2.$$

Finally, $(s,t) = (2,2m)$ gives (7.7); Parseval's identity applied to $p^2$ gives (7.8).

For $p(z) = c(z^n - \omega)$, rotation invariance gives $\|p\|_r = |c| C_r$ for every $0 \le r \le \infty$. Hence all constants asserted in the corollary are attained. ☐

7.2. **Erdős–Szekeres products.** Applying the $L^2$-to-$L^\infty$ comparison to a product of cyclotomic factors gives the following consequence.

**Corollary 7.2.** *Let $N \ge 1$ and let $s_1, \dots, s_N$ be positive integers. Define $P_N(z) := \prod_{j=1}^{N}(1 - z^{s_j}) = \sum_k b_k z^k$. Then*

$$\|P_N\|_\infty^2 \ge 2\sum_k |b_k|^2, \qquad \sum_k |b_k|^2 \ge 2N, \tag{7.9}$$

*and therefore*

$$\|P_N\|_\infty \ge 2\sqrt{N}. \tag{7.10}$$

*Proof.* Every zero of $P_N$ lies on $\mathbb{T}$, so the first inequality in (7.9) is (7.5). Moreover, $zP_N(z) \in \mathbb{Z}[z]$ has zero constant term and is divisible by $(1-z)^N$. Tang's theorem [Tan26, pp. 3384–3385, Theorem 3.2], applied to $zP_N$, gives

$$\sum_k \left| [z^k]\big(zP_N(z)\big) \right|^2 \ge 2N,$$

where $[z^k]F(z)$ denotes the coefficient of $z^k$ in the polynomial $F(z)$. Multiplication by $z$ only shifts the coefficient sequence, so

$$\sum_k \left|[z^k]\big(zP_N(z)\big)\right|^2 = \sum_k |b_k|^2.$$

Combining this with the first inequality in (7.9) proves (7.10). □

7.3. **A mixed derivative inequality.** The extremal derivative identity for unimodular-zero polynomials combines with Theorem 1.1 as follows.

**Corollary 7.3.** *Let $p$ be as in (7.2), and let $C_r$ be defined by (7.1). Then, for every $0 \le r \le \infty$,*

$$\|p'\|_\infty = \frac{n}{2}\|p\|_\infty \ge \frac{n}{C_r}\|p\|_r . \tag{7.11}$$

*The constant $n/C_r$ is sharp. In particular,*

$$\|p'\|_\infty \ge n|a_n|, \qquad \|p'\|_\infty \ge \frac{\pi n}{4}\|p\|_1, \qquad \|p'\|_\infty \ge \frac{n}{\sqrt{2}}\|p\|_2 .$$

*Proof.* Since $p$ has no zero in $\mathbb{D}$, the Erdős–Lax theorem gives $\|p'\|_\infty \le (n/2)\|p\|_\infty$ [Lax44, pp. 509–513]. Since all zeros of $p$ lie in $\overline{\mathbb{D}}$, Turán's reverse inequality gives the opposite bound [Tur40, p. 90, Satz I]. Hence equality holds in the first part of (7.11); see also [OR74, p. 332, Theorem 1]. The second part follows from (7.3) with $t = \infty$. For $p(z) = c(z^n - \omega)$, both inequalities are equalities, so the constant is sharp. □

7.4. **Logarithmic variance and power sums.** For the zeros in (7.2), define

$$S_k := \sum_{j=1}^{n} \zeta_j^k, \qquad k \ge 1. \tag{7.12}$$

The behavior of the norm ratio at the zero exponent gives a sharp logarithmic variance bound.

**Corollary 7.4.** *Let $p$ and $S_k$ be defined by (7.2) and (7.12). Then*

$$\int_{\mathbb{T}} \big(\log|p| - \log|a_n|\big)^2 \, dm \ge \frac{\pi^2}{12}, \tag{7.13}$$

*and equivalently*

$$\sum_{k=1}^{\infty} \frac{|S_k|^2}{k^2} \ge \frac{\pi^2}{6}. \tag{7.14}$$

*Equality holds when the zeros form a regular $n$-gon.*

*Proof.* For a nonzero polynomial $f$, define $Y_f(\zeta) := \log|f(\zeta)|$ when $f(\zeta) \ne 0$, and set $Y_f(\zeta) := 0$ on its finite zero set. Then $Y_f \in L^q(\mathbb{T}, m)$ for every finite $q$, its positive part is bounded, and $e^{rY_f} = |f|^r$ almost everywhere. On $\{Y_f < 0\}$ and for $0 < r \le 1$,

$$0 \le e^{rY_f} - 1 - rY_f \le \frac{r^2 Y_f^2}{2};$$

on $\{Y_f \ge 0\}$ the same remainder is bounded by $C_f r^2 Y_f^2$, because $Y_f^+$ is bounded. The dominated convergence theorem therefore yields

$$\int_{\mathbb{T}} e^{rY_f} \, dm = 1 + r\int_{\mathbb{T}} Y_f \, dm + \frac{r^2}{2}\int_{\mathbb{T}} Y_f^2 \, dm + o(r^2).$$

Taking the logarithm and dividing by $r$ gives

$$\log\|f\|_r = \int_{\mathbb{T}} Y_f \, dm + \frac{r}{2}\operatorname{Var}_m(Y_f) + o(r) \qquad (r \downarrow 0), \tag{7.15}$$

where $\mathrm{Var}_m(Y) := \int_{\mathbb{T}} Y^2 \,\mathrm{d}m - \left(\int_{\mathbb{T}} Y \,\mathrm{d}m\right)^2$ denotes the variance with respect to $m$. Set

$$F(r) := \log \|p\|_r - \log \|Q_n\|_r .$$

By Theorem 1.1, $F(r) \geq F(0)$ for every $r > 0$. Equation (7.15) shows that the right derivative $F'(0+)$ exists; dividing $F(r) - F(0) \geq 0$ by $r$ and letting $r \downarrow 0$ gives $F'(0+) \geq 0$. Hence

$$\mathrm{Var}_m(\log |p|) \geq \mathrm{Var}_m(\log |Q_n|).$$

Jensen's formula gives $\int \log |p| \,\mathrm{d}m = \log |a_n|$, while

$$\log |1 + \mathrm{e}^{i\theta}| = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} \cos(k\theta) \quad \text{in } L^2(0, 2\pi).$$

Because $z \mapsto z^n$ preserves the normalized Haar measure $m$, the variance of $\log |Q_n|$ equals that of $\log |1 + z|$. Parseval's identity therefore gives $\mathrm{Var}_m(\log |Q_n|) = \frac{1}{2} \sum_{k \geq 1} k^{-2} = \pi^2/12$, proving (7.13).

For each unimodular $\zeta_j$,

$$\log |\mathrm{e}^{i\theta} - \zeta_j| = -\sum_{k=1}^{\infty} \frac{1}{k} \mathrm{Re}\big(\overline{\zeta_j}^{\,k} \mathrm{e}^{ik\theta}\big) \quad \text{in } L^2(0, 2\pi).$$

Summing over $j$ and applying Parseval gives

$$\int_{\mathbb{T}} \big(\log |p| - \log |a_n|\big)^2 \,\mathrm{d}m = \frac{1}{2} \sum_{k=1}^{\infty} \frac{|S_k|^2}{k^2}.$$

Thus (7.14) is equivalent to (7.13). If the zeros form a regular $n$-gon, then $S_k = 0$ unless $n \mid k$, while $|S_{jn}| = n$; equality follows. □

7.5. **Relative entropy at every exponent.** Differentiating the monotone norm ratio at a positive exponent recovers the entropy comparison itself.

**Corollary 7.5.** *Let $p$ be as in (7.2), and set $Q_n(z) := 1 + z^n$. For $r > 0$ and $f \in \{p, Q_n\}$, define $\rho_{f,r}$ by (2.9). Then*

$$\mathsf{D}(\rho_{p,r} \| m) \geq \mathsf{D}(\rho_{Q_n,r} \| m), \qquad r > 0. \tag{7.16}$$

*Equality is attained by $p(z) = c(z^n - \omega)$ with $c \neq 0$ and $|\omega| = 1$.*

*Proof.* The function $r \mapsto \log \|p\|_r - \log \|Q_n\|_r$ is differentiable for $r > 0$ and nondecreasing by Theorem 1.1. Its derivative is therefore nonnegative. Lemma 2.3 identifies $r^2$ times that derivative with the left-hand side minus the right-hand side of (7.16).

For $p(z) = c(z^n - \omega)$, multiplication by $c$ cancels in the normalized density. Choose $\eta \in \mathbb{T}$ with $\eta^n = -\omega$; then $|(\eta z)^n - \omega| = |1 + z^n|$. Rotation invariance therefore shows that $\rho_{p,r}$ and $\rho_{Q_n,r}$ have the same entropy, proving the equality assertion. □

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